A validity criterion and a usage map for likelihoods of macroscopic ion-channel currents

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Abstract Macroscopic ion-channel recordings carry stochastic fluctuations that the usual least-squares analysis discards. Likelihoods that use them exist, including Bayesian recursive filters, but are rarely applied: unlike a least-squares fit, which can be judged from its residuals, a recursive filter tracks the data whether or not it reports its uncertainty correctly, and whether it does has never been characterised. Here we test a ladder of likelihoods, from least squares to a filter conditioned on both ends of the acquisition interval, against data simulated exactly from the process they approximate. At the true parameters the score must average to zero and its covariance must equal the Fisher information the likelihood reports; we measure the departure in each, which is the bias and the distortion respectively. A likelihood that models the gating variance but carries no recursion is calibrated interval by interval and under-reports parameter uncertainty ten to fifteen fold once a record accumulates; a sandwich correction restores nominal coverage. The boundary-conditioned filter reports honestly at 94% of the measured points, departing toward few channels. The region where the fluctuations still determine the amplitude parameters and the region where the classical error bar is honest never overlap, lying one to three decades of noise apart. This supplies a test of whether a macroscopic-current filter tells the truth about its own uncertainty, and a map of where it does not.

Introduction

A macroscopic ion-channel current is the summed signal of a population of channels. Its mean carries the gating kinetics through the deterministic relaxation, and its fluctuations carry, in addition, the number of channels, the unitary current, and, through the temporal structure of the noise, the rate constants themselves. The standard way to fit a kinetic scheme to such a recording is least squares on the mean current. The four approaches surveyed by **?** are all deterministic fits with a root-mean-square objective, and the community benchmark IonBench states that it does not cover optimisation approaches for stochastic models (**?**).

Least squares rests on an assumption that a gating record does not satisfy. The residual sum of squares becomes a variance, and then a confidence interval, only through a count of independent residuals. Channel gating leaves the residuals correlated over the channel's own relaxation time, so a sample taken faster than that time partly repeats what the sample before it already said. The point estimate often survives this. The count does not: a fit that treats correlated residuals as independent behaves as though the recording supplied more observations than it did, and reports an interval narrower than the spread of the estimator it is describing. **?** state the same cost in the field's own terms, that standard methods built for deterministic models do not distinguish stochastic channel gating from measurement noise and return biased estimates.

A Markov description of gating models the dependence instead of assuming it away. A channel with a small number of conformational states, connected by rate constants that hold over the whole record, generates the mean current and the covariance of the fluctuations from the same constants. The temporal structure of the noise then becomes a second measurement of the same parameters rather than a nuisance to be neutralised. The rate constants also mean something outside the record: they are properties of the protein, comparable across conditions, preparations and laboratories, and they connect what is measured to a mechanism. A correlation function fitted freely to the residuals delivers neither of those.

Using the fluctuations to identify a scheme requires a likelihood for the recorded current, and the likelihoods that exist for macroscopic recordings form a small lineage, each of them an approximation. **?** fit the full Q-matrix covariance directly, at a cost that grows as the cube of the record length. **?** keeps the gating variance at each sample and drops the correlation between samples. **?** recasts the problem as a recursive filter, propagating the occupancy distribution and conditioning it on each new datum. **?** restores the full covariance at near-linear cost through a semiseparable representation, and **?** places the recursive filter in a Bayesian setting. A recent member conditions the interval-averaged observable on the channel states at both ends of the acquisition interval (MacroIR, **?**). Ordered by what they compute for every interval of the record, these methods form a ladder of cost with least squares at its bottom rung, and the ladder hangs from one root question: does the analysis model the gating fluctuations at all, and if it does, how much of their temporal structure does it keep?

None of these methods has displaced least squares. They are published, implemented and available, and they are almost unused. The reason has to do with what an analyst can see.

Least squares is checkable by eye. It predicts the current at every point of the record from the parameters alone, so the prediction can be laid over the data and read interval by interval. A systematic departure of the model from the record shows up as a systematic departure of the line from the points. When the scheme is wrong the picture usually says so, and when the forward model has a mistake in it the picture usually says that too. This is weak evidence in a formal sense and strong evidence in practice, and it is what has kept the classical fit in place.

A recursive filter offers no such picture. It predicts each interval from the parameters and from the data already observed, so the previous sample enters the prediction of the next one and the predicted trace follows the record closely by construction. It follows the record closely when the kinetic scheme is right, when the kinetic scheme is wrong, and when the code has a mistake in it. The residuals come out small and close to white, and they come out white in large part because the recursion subtracts, at each step, what it has just seen. Whiteness of the residuals is as far as the prior work goes, and it cannot falsify anything, since enough free terms will whiten any sequence. The visual evidence that carries the classical fit is unavailable for a filter, and nothing has taken its place. There has been no straightforward way to tell a working method from a bug, and no criterion for deciding whether the confidence interval it reports, which is the reason to use a likelihood in the first place, should be believed. Supplying that criterion is what this paper is for.

The standard statistical repair for correlated residuals does not reach this problem. Generalized least squares with autoregressive or moving-average errors whitens the residual sequence with a correlation function estimated from the residuals themselves, and **?** have run exactly that on ion-channel data, fitting an autoregressive moving-average error model and two Gaussian-process variants to the residuals of a hERG kinetic model. Gating puts this family out of reach for structural reasons. An autoregressive moving-average error model is stationary by construction, one covariance function for the whole trace, while the gating covariance tracks the mean current, vanishes when the current does, and restarts at every concentration or voltage jump. It contains no channel number and no unitary current, so it cannot return the two parameters the fluctuations are richest in. Its timescales are free parameters, where the Markov model ties them to the eigenvalues of the rate matrix that already fixed the mean, so free timescales absorb the information the mechanism would have contributed. **?** report that the autoregressive model widens the posterior without predicting better, and that in some cases the best predictions came from ignoring the discrepancy altogether. Their target is structural error in the kinetic scheme rather than finite-channel gating noise, so their study is not a test of the present question; what it shows is what a correlation model without a mechanism can and cannot buy.

None of this is a claim that the correlation went unnoticed. Kinetic information has been read out of the temporal correlation of macroscopic currents since the early 1970s, first through the Lorentzian power spectrum under stationary conditions (**???**), which is the Fourier transform of the autocovariance. **?** obtained the unitary conductance and the channel closing rate jointly, from the same fluctuation data and with standard errors on both: a unitary conductance of 0.205 ± 0.0063 (standard error) $\times 10^{-10}$ mho, and a closing rate exponential in membrane potential whose prefactor is 0.17 ± 0.04 ms^{-1}. The exact non-stationary two-time covariance for a population of identical independent Markov channels was derived and used a decade later (**??**), estimated from ensembles of 256 to 504 repeated sweeps, from which rate constants were obtained with standard errors, and used principally to discriminate between kinetic hypotheses. A covariance likelihood for macroscopic currents (**?**) predates the recursive filter of **?** by three years. The statistical apparatus used below is equally old: the score, the information matrix equality and the sandwich covariance under misspecification are classical (**???**). What none of this work supplies is a statement of when a method built on the correlation is working.

Non-stationary fluctuation analysis is the one fluctuation-based method in routine use. It regresses the variance of the current against its mean and returns the unitary current, the channel number and the peak open probability (**?**), and practitioners of the method report that the unitary current is close to the only parameter it recovers reliably and that kinetic rates have essentially never been estimated this way for receptors in their native setting. This paper does not compete with it over how much amplitude information a macroscopic recording holds. The question here is whether a reported confidence interval from a fit to such a recording is honest, and where, on the plane of channel number against instrumental noise, it stops being honest. Nothing in the fluctuation-analysis literature addresses that. What the classical arm of this paper is should also be said plainly: least squares on the deterministic mean current with the unitary current held fixed. Fluctuation analysis is never benchmarked here, and no comparison against the variance-mean regression should be read into any result below.

The criterion this paper supplies is checkable. Under a correctly specified likelihood the score, the gradient of the log-likelihood with respect to the parameters, averages to zero at the true parameters, and the covariance of the score equals the Fisher information the likelihood reports, which is the information matrix equality (**??**). Under an approximation both statements fail, and how they fail is measurable. Goodness of fit does not answer the question, because an approximation can reproduce the mean current closely and still report an information matrix wrong by more than an order of magnitude, and comparing two approximations to each other does not answer it either, because both may be wrong. What makes the question answerable for this class of models is an asymmetry between simulation and inference: the forward process can be simulated exactly while its likelihood cannot be computed exactly, so the simulator supplies the ground truth that the likelihood is an approximation of, and any candidate likelihood can be interrogated against data drawn from the very process it claims to describe. The sensitivity matrix is then analytic and the variability matrix is a Monte Carlo average over independent replicates, with no null hypothesis to reject and no finite-sample defect to correct for. This is measurement rather than testing.

The mismatch between the two matrices is itself the measurement. It gives an information distortion matrix that says, in units of parameter variance a reader already understands, by how much and in which parameter directions an approximation misreports what the data determine. The distortion runs in both directions: some approximations state more confidence than the data support, and others state less. Evaluated across channel number and instrumental noise at a fixed acquisition interval, the same quantity yields a map of which parameters a macroscopic recording can deliver and whose reported interval is honest, with boundaries drawn as level sets of a continuous diagnostic rather than as thresholds. The map carries an ordering that is falsifiable and does not depend on where those level sets are placed: there is no region of the measured plane in which the amplitude still carries information and the classical error bar is simultaneously trustworthy, the two boundaries being separated by one to three decades of instrumental noise everywhere they were measured. Wherever the fluctuations still say something about the unitary current or the channel number, a classical fit reports an interval that is too narrow.

Against that history, what is new here is narrow and is worth stating narrowly. A joint estimate of rates and amplitudes from fluctuation data, carrying standard errors, is **?**. What this paper adds, to a non-stationary protocol read from a single record, is the third element of the combination: an interval whose calibration has been verified rather than assumed. The thousands of replicate recordings used below certify the method; they are not what applying it requires.

The measurements are made in a deliberately small setting: a two-state channel whose open probability at the peak of the response is fixed at one half, a single concentration jump, simulated data, and comparison at the level of the likelihood rather than of a fitted experiment. The small model isolates the error the approximation itself commits, with no confounding from a misspecified kinetic scheme, and it keeps the exact simulation cheap enough to run thousands of replicates at every design cell. Richer schemes, the stationary regime, the few-channel boundary and experimental recordings are the subjects of companion work.

The map answers a question the field is currently arguing about. **?** argue against filtering on cost, since the moment integration and the correction step run between every pair of samples, and they are right about the cost. Knowing which approximation misreports the uncertainty, by how much, and in which corner of the operating regime is what turns that cost into a decision a reader can make for their own rig.

The macroscopic interval likelihood and its approximations

A macroscopic current is the summed signal of a population of ion channels, each of which behaves as an independent continuous-time Markov chain over a small set of conformational states. We write N_{ch} for the number of channels and K for the number of kinetic states of one channel ($K = 2$ throughout this paper, carried symbolically where it costs nothing). Each channel is governed by a generator (rate) matrix $\mathbf{Q} \in \mathbb{R}^{K\times K}$, whose transition matrix over an elapsed time t is the row-stochastic $\mathbf{P}(t) = e^{\mathbf{Q}t}$ with entries $P_{i\to j}(t)$. A channel in state k carries a conductance γ_k, and we collect these in the vector $\boldsymbol{\gamma} \in \mathbb{R}^K$.

The likelihood we need is the probability of a recorded sample given the kinetic parameters and the data seen so far. Its exact form would require the distribution of the interval-averaged conductance of the whole population, which has no closed expression. Every practical likelihood replaces that distribution with a Gaussian, and the members of the family studied here differ in what the Gaussian is conditioned on and in how its variance is accounted. The rest of this section unpacks that sentence.

The observable is an interval average

A recorded sample is the current averaged over the acquisition window rather than sampled at an instant. The acquisition electronics integrate the current across the sampling window Δ and pass it through an anti-aliasing filter before it is written down, so the quantity a likelihood must describe is the windowed average

$$\bar{y}_t \;=\; \frac{1}{\Delta}\int_{t-\Delta}^{t} y(\tau)\,\mathrm{d}\tau \;+\; \eta_t, \tag{1}$$

where $y(\tau)$ is the instantaneous population current and η_t is the instrumental noise accumulated over the window. This interval average is the physical observable. When Δ is much shorter than the fastest relaxation of $\mathbf{Q}$ the average sits close to the instantaneous current and the distinction is negligible; when Δ is comparable to or longer than that relaxation the two diverge, and a likelihood built on instantaneous sampling describes the wrong random variable.

Two Gaussian approximations

Each member of the family makes two closures, on two different objects.

The two Gaussian approximations

Occupancy closure. The joint distribution of the channel occupancies is exactly multinomial when the channels are independent, which is also the maximum-entropy distribution consistent with the mean occupancies. It is replaced by a multivariate Gaussian with mean $\boldsymbol{\mu}$ and covariance $\boldsymbol{\Sigma}$, valid for large N_{ch} by the central limit theorem and degrading as the channel count falls.

Interval-signal closure. The distribution of the interval-averaged conductance over one window is a telegraph-like object with no closed form. It is replaced by a Gaussian, valid when the window spans many transitions or when instrumental noise dominates, and degrading as the window shrinks below the relaxation time.

The higher moments these closures discard do not vanish. They reappear in the data as temporal correlation that the likelihood does not predict, which is what the correlation term of the information-distortion decomposition measures (Diagnostics). Three regimes follow, and we name them here so the maps in the Results have vocabulary: the *multinomial* regime (few channels, where the occupancy closure fails), the *telegraphic* regime (windows far shorter than the relaxation time, where the interval-signal closure fails), and the *Gaussian* regime, where both closures hold because the channel count is high enough for the occupancy Gaussian while the window is long enough, or the instrumental noise large enough, for the interval-signal Gaussian. Any Gaussian likelihood of this kind must fail somewhere. The aim of this paper is to show that the failures are the predicted degradation of these two closures rather than defects of a particular algorithm.

The usage map of the Results is drawn on the plane of channel count against instrumental noise at one acquisition interval, and its region names follow this vocabulary. Region 0, where the closure itself is misspecified, is the multinomial regime seen on that plane, and its boundary carries a negative slope because instrumental noise makes the emission more Gaussian and so relaxes the occupancy closure. Regions 1 to 4 (the unitary current and channel count with a calibrated interval; the rates with a calibrated interval; least squares at par; nothing estimable) all sit inside the Gaussian regime, and what orders them is how much the data carry rather than whether the closures hold. The telegraphic regime is a statement about Δ and so does not appear as a region of that plane; it is reached by moving along the interval axis of Figure **??**.

The root question, and the ladder below it

Before any of the choices above comes a prior question: does the method model the gating fluctuations at all? Classical nonlinear least squares (LSE) answers no. It fits the deterministic mean current and assigns one constant variance to every residual, so it carries neither an occupancy distribution nor a conductance conditioned on anything, and the temporal structure the two closures approximate is absent from it. LSE is therefore not a member of the family. It is the bottom rung of the ladder of computational cost this paper walks, and it is the anchor against which every member above it is measured, because it is the method most readers are using.

Given a yes to the root question, the members differ along two axes. One axis is the window: how much of the acquisition interval the conductance is conditioned on. The other is recursion: whether the occupancy distribution is conditioned on the data as the recording proceeds.

Along the recursion axis, a recursive (prequential) member factorizes the likelihood as $\prod_t q_t(\bar{y}_t \mid \bar{y}_{1:t-1})$, so the occupancy distribution carried into each window is the posterior left by the previous windows and the hidden state is conditioned on the data. A non-recursive (open-loop) member factorizes as $\prod_t m_t(\bar{y}_t)$, with each window's predictive propagated from the initial condition and never conditioned on the observed trace.

Along the window axis the members form a ladder, indexed by how many of the interval's endpoints the conductance is conditioned on (Table **??**).

Conditioned on	Members	Conductance model
the fluctuations are not modelled	`LSE`	the deterministic mean current only
no endpoints	`NR`, `R`	instantaneous; the acquisition averaging is ignored
one endpoint (the start)	`MR`, `VR`	interval-mean conductance given the start state
two endpoints (the boundary)	`IR`	interval-mean conductance given both boundary states; i
the full trajectory	(exact)	intractable; supplied by the stochastic simulation as grou

Table 1. The window axis as an endpoint ladder, with least squares (`LSE`) shown at the foot of the cost ladder although it lies off the lattice. Crossing the window axis with the recursion axis gives the implemented grid: `NR` (non-recursive, $\mathrm{av} = 0$), `R` (recursive, $\mathrm{av} = 0$), `MR` (recursive, $\mathrm{av} = 1$, total start-conditioned interval variance), `VR` (recursive, $\mathrm{av} = 1$, residual interval variance), `IR` (recursive, $\mathrm{av} = 2$), where av counts the conditioned endpoints. The body walks `LSE`, `NR`, `R`, `IR`; `MR` and `VR` split the step from `R` to `IR` and are reported in the Supplement. MacroIR is `IR`.

Two things follow from the ladder. First, `IR` occupies the top rung below the intractable exact likelihood, so its standing in the family is structural and set before any measurement. Second, the lower rungs are established likelihoods rather than constructed foils: the non-recursive instantaneous member is the independent-interval likelihood of **?**, and the recursive instantaneous member is the macroscopic recursive filter of **?** and the generalized filter of **?**. The lattice is one coordinate system that contains these methods as special cases.

The boundary state

A *boundary state* is the pair (i_0, i_t) of a channel's states at the two ends of an acquisition interval. Chaining intervals, the end state of one interval is the start state of the next, so there is a single state variable per junction: the filter conditions on the states at the interval boundaries and marginalizes the trajectory between them analytically. This is the same device as static condensation, or the spatial Markov property, applied on the time axis. It is not a transition state in the mechanistic sense, and the word *transition* is not used for it here. Because the joint dynamics of the two endpoints are generated by the Kronecker sum $\mathbf{Q} \oplus \mathbf{Q} = \mathbf{Q} \otimes \mathbf{I} + \mathbf{I} \otimes \mathbf{Q}$, whose propagator factorizes as $e^{(\mathbf{Q}\oplus\mathbf{Q})t} = e^{\mathbf{Q}t} \otimes e^{\mathbf{Q}t}$, the moments over the K^2 boundary pairs collapse to bilinear forms in the K-dimensional state space and no explicit K^2 object is formed (Methods).

The ladder needs two conductance objects, both built from the boundary-conditioned single-channel moments $\overline{\Gamma}_{i_0 \to i_t}$ (the mean interval-averaged conductance of a channel that starts in i_0 and ends in i_t) and $\overline{V}_{i_0 \to i_t}$ (its variance), which are computed from $\mathbf{Q}$ and the recording filter (Methods). The start-conditioned mean conductance is the K-vector

$$(\bar{\gamma}_0)_{i_0} \;=\; \sum_{i_t} P_{i_0 \to i_t}(\Delta)\, \overline{\Gamma}_{i_0 \to i_t}, \tag{2}$$

which conditions on the state at the interval's start alone and is the object used at $\mathrm{av} = 1$. The boundary-conditioned mean conductance is the same average retained on both endpoints, so it stays a $K \times K$ object rather than collapsing to a vector, and is the object used at $\mathrm{av} = 2$.

The interval update

Over a window of length Δ, the occupancy mean and covariance propagate as

$$\boldsymbol{\mu}^{\mathrm{prop}} = \mathbf{P}(\Delta)^{\top} \boldsymbol{\mu}_0, \tag{3}$$

$$\boldsymbol{\Sigma}^{\mathrm{prop}} = \mathbf{P}(\Delta)^{\top} \big(\boldsymbol{\Sigma}_0 - \mathrm{diag}(\boldsymbol{\mu}_0)\big) \mathbf{P}(\Delta) + \mathrm{diag}(\boldsymbol{\mu}^{\mathrm{prop}}), \tag{4}$$

where $(\boldsymbol{\mu}_0, \boldsymbol{\Sigma}_0)$ is the occupancy Gaussian at the start of the window. The predicted mean of the interval-averaged current is

$$\bar{y}^{\mathrm{pred}} \;=\; N_{\mathrm{ch}}\, \boldsymbol{\mu}_0^{\top} \bar{\boldsymbol{\gamma}}_0, \tag{5}$$

and its predicted variance decomposes into three independent contributions,

$$\sigma^2_{\mathrm{pred}} \;=\; \epsilon^2_{\Delta} \;+\; N_{\mathrm{ch}}\, \widetilde{\boldsymbol{\gamma}^{\top}\boldsymbol{\Sigma}\boldsymbol{\gamma}} \;+\; N_{\mathrm{ch}}\, \boldsymbol{\mu}_0^{\top} \boldsymbol{\sigma}^2_{\bar{\gamma}_0}. \tag{6}$$

The first term ϵ^2_{Δ} is the instrumental noise over the window, which carries the acquisition averaging and decreases with Δ; its parametric form is given in Methods. The second term is the population gating variance, the fluctuation of the window current transmitted through the occupancy covariance: $\widetilde{\boldsymbol{\gamma}^{\top}\boldsymbol{\Sigma}\boldsymbol{\gamma}}$ is the boundary covariance of the window collapsed onto the conductance, computed without ever forming the K^2 boundary matrix. The third term is the within-window per-channel variance, with $\boldsymbol{\sigma}^2_{\bar{\gamma}_0}$ the start-conditioned interval variance built from $\overline{V}_{i_0 \to i_t}$.

Writing $\mathbf{g} = \mathrm{Cov}(\text{end-of-window occupancy}, \bar{y})/N_{\mathrm{ch}}$ for the per-channel cross-covariance between the interval current and the occupancy at the end of the window, and $\delta = \bar{y}^{\mathrm{obs}} - \bar{y}^{\mathrm{pred}}$ for the innovation, the Gaussian update of the occupancy is the scalar Kalman correction

$$\boldsymbol{\mu}^{\mathrm{post}} = \boldsymbol{\mu}^{\mathrm{prop}} + \frac{\mathbf{g}}{\sigma^2_{\mathrm{pred}}}\,\delta, \tag{7}$$

$$\boldsymbol{\Sigma}^{\mathrm{post}} = \boldsymbol{\Sigma}^{\mathrm{prop}} - \frac{N_{\mathrm{ch}}\,\mathbf{g}\mathbf{g}^{\top}}{\sigma^2_{\mathrm{pred}}}. \tag{8}$$

The factor N_{ch} enters the covariance downdate but not the mean correction because $\boldsymbol{\Sigma}$ is carried on the population scale, the same N_{ch} that multiplies the gating variance in Eq. **??**, whereas $\boldsymbol{\mu}$ and the gain $\mathbf{g}$ are per-channel objects, so the rank-one correction to $\boldsymbol{\Sigma}$ must be rescaled by N_{ch} to sit on its scale while the correction to $\boldsymbol{\mu}$ needs no such factor. In a recursive member the posterior pair $(\boldsymbol{\mu}^{\mathrm{post}}, \boldsymbol{\Sigma}^{\mathrm{post}})$ becomes the prior for the next window; in a non-recursive member it is not carried forward, and each window's predictive is propagated from the initial condition alone. Reusing the posterior as the next prior is itself a closure: it is exact only while the end marginal is a sufficient summary of the boundary posterior, which the Gaussian projection does not guarantee. For this reason the recursive members' Gaussian information is a model-based quantity and is not claimed to be the exact information of the process; how far it departs from the truth is what the Diagnostics measure.

Two algebraic differences separate the members that sit at $\mathrm{av} = 1$ and $\mathrm{av} = 2$. The first is the form of the third term of Eq. **??**. `MR` carries the total per-start-state interval variance, the spread of the interval-averaged conductance of a channel known only to have started in i_0, while `VR` and `IR` carry the residual variance left once the end state is also conditioned on, $\sum_{i_t} P_{i_0 \to i_t}(\Delta)\, \overline{V}_{i_0 \to i_t}$. By the law of total variance the two forms differ by the spread of the interval mean across end states,

$$\underbrace{\sum_{i_t} P_{i_0\to i_t}(\Delta)\Big[\overline{V}_{i_0\to i_t} + \big(\overline{\Gamma}_{i_0\to i_t} - (\bar{\gamma}_0)_{i_0}\big)^2\Big]}_{\text{total, MR}} \;=\; \underbrace{\sum_{i_t} P_{i_0\to i_t}(\Delta)\,\overline{V}_{i_0\to i_t}}_{\text{residual, VR, IR}} + \mathrm{Var}_{i_t}\big[\overline{\Gamma}_{i_0\to i_t} \mid i_0\big], \tag{9}$$

so `MR` charges to the observation variance a spread that the end state, once conditioned on, would resolve. The second difference is the boundary cross-covariance term

$$N_{\mathrm{ch}}\, \widetilde{\boldsymbol{\gamma}^{\top}\boldsymbol{\Sigma}\boldsymbol{\gamma}} \tag{10}$$

that enters the gain $\mathbf{g}$ of Eqs. **??** and **??**, which `IR` retains and both $\mathrm{av} = 1$ members drop. Stepping `MR` to `VR` changes the variance form and nothing else, and stepping `VR` to `IR` restores the boundary term in the gain and changes nothing else, so the two halves of the `R` to `IR` step are separated and can be measured one at a time. Their consequences for the predicted variance across the regime grid are characterized in the Results.

The top of the ladder

The exact likelihood conditions on the full trajectory inside each window and is intractable, which is why the two closures above are needed at all. The same stochastic model that makes the exact likelihood intractable can be simulated forward exactly, one trajectory at a time, so the simulation realizes the object at the top of the ladder and serves as the ground truth against which every rung below it is judged.

Diagnostics: testing a likelihood against its simulation

An approximate likelihood is a misspecified model, and misspecification leaves a classical signature. At the true parameters the score no longer has zero mean, so the estimator the likelihood defines is biased; and the covariance of the score no longer equals the negative expected curvature, so the uncertainty the likelihood reports is wrong. These are the two Bartlett identities, and their failure is the content of the information-matrix-equality diagnostic of **?** and of the sandwich covariance of **??**. In the classical setting those identities can only be tested, through a statistic with a known finite-sample defect, because the data-generating law is unknown. Our setting removes that limitation. The forward process is an exact continuous-time Markov chain (CTMC) and we simulate it exactly, so we hold unlimited data drawn from the very law the likelihood claims to describe, at parameters we set. The classical test therefore becomes a direct measurement: the anchor curvature is computed analytically and the score covariance is estimated by Monte Carlo over independent simulated recordings, with no null hypothesis to reject. The apparatus is standard; what is new here is applying it in this domain and reading its output as a measured property of each approximation rather than as a hypothesis to accept or reject.

We read three quantities off the simulation, in increasing strength. The predictive residual and the information anchor are evaluated at the estimate; the expectation behind the score bias is taken at the simulation truth. For a misspecified likelihood the optimum and the truth differ, and that optimum-versus-truth gap is what makes the bias observable.

The first is the standardized residual. With μ_t and σ_t^2 the likelihood's own predictive mean and variance for interval t, the residual $r_t = (y_t - \mu_t)/\sigma_t$ must have mean zero, unit variance, and no autocorrelation across intervals if the one-step predictive distribution is correct (Figure **??**). This probes the predictive one interval at a time. It is the cheapest of the three and the only one an experimentalist can run on a real recording, since it needs the predictive moments but not the true parameters. Residual whiteness in particular is not optional: residual autocorrelation is the same temporal signal that the strongest check quantifies below.

The second is the score. Writing $\ell(\theta)$ for the total log-likelihood and $S(\theta) = \nabla_\theta \ell(\theta)$ for its gradient, correct specification requires $\mathbb{E}[S(\theta^*)] = 0$ at the true parameters θ^*. A nonzero mean score is what biases the estimate, and projected through the parameter covariance it becomes a displacement in parameter units, $b = \mathbf{H}^{-1}\,\mathbb{E}[S]$, the reader-facing statement of how far the estimate is pushed (Figure **??**). This projection is a first-order approximation, from the expansion of the score at the optimum given below. Unlike the residual, the check needs a known truth, so it requires the simulator, and its confidence interval narrows as the number of simulated recordings grows.

The third and strongest is the information itself. Correct specification requires $\mathrm{Cov}[S] = \mathbf{H}$, the information the likelihood reports. An approximation can pass the residual and score checks and still violate this one by an order of magnitude. The two failures are independent: whether the estimate is biased is governed by how the conductance is modelled over the interval, while whether the reported information agrees with the information delivered is governed by the recursion. We turn all three checks on the whole cost ladder set out in Theory, from least squares on the deterministic mean current (LSE) at the bottom to the boundary-conditioned filter at the top. The range they have to cover is wide. The members that carry no recursion misreport the information grossly, and among the recursive members the interval conditioning leaves distortions small enough that a diagnostic has to be sharp to resolve them, so the same instrument has to remain informative across both. Because the failure is directional, we record it as a matrix,

$$\mathbf{C} = \mathbf{H}^{-1/2}\,\mathbf{J}\,\mathbf{H}^{-1/2},$$

with $\mathbf{J} = \mathrm{Cov}(S)$ the covariance of the total score, cross-interval terms included. This definition is exact algebra given $\mathbf{H}$ and $\mathbf{J}$, and under correct specification $\mathbf{J} = \mathbf{H}$ so $\mathbf{C} = \mathbf{I}$ exactly. We fix the direction by convention: $C_{ii} > 1$ means the likelihood under-reports the uncertainty in parameter i and is over-confident, $C_{ii} < 1$ means it over-reports and is conservative. The eigen-directions of $\mathbf{C}$ say which combinations of parameters are misreported, which a scalar summary cannot recover.

The anchor $\mathbf{H}$ is the likelihood's own Gaussian Fisher information, $\mathbf{H} = \sum_t \mathbb{E}\big[(\nabla\mu_t)(\nabla\mu_t)^\top/\sigma_t^2 + (\nabla\sigma_t^2)(\nabla\sigma_t^2)^\top/($ formed from the same predictive moments and their parameter derivatives that the score already requires, at no extra cost. For a macroscopic algorithm the likelihood is Gaussian by construction, so this is the model's own Fisher and $\mathbf{C}$ is the information-matrix-equality test carried out in the model's own frame. Two properties follow. The Gaussian Fisher is positive semidefinite by construction, so the diagnostic never has to be rescued from a numerically indefinite anchor. And it is analytic, so it does not inherit the finite-difference noise of a numerical Hessian, which in this regime is severe: the finite-difference Fisher is widely indefinite per replicate, and only pooling across replicates recovers a definite matrix. By design the anchor is the Gaussian Fisher; the numerical Fisher we compute alongside it serves to gauge how faithful that Gaussian anchor is. Comparing the approximation to its own Fisher is not circular. $\mathbf{H}$ is the information the likelihood claims, $\mathbf{J}$ is the information it delivers against data from the true process, the two are computed from different objects, and their disagreement is the definition of misspecification.

The bottom rung of the ladder needs a qualification at the point where the diagnostic is defined, because its predictive variance is not a gating quantity. Least squares fits the deterministic mean current under a single variance held constant over the whole recording, so the information it reports, $\mathbf{H}$ as written above with σ_t^2 replaced by one constant σ^2, is the information of a homoscedastic model. The exact process violates that assumption: the variance of a recorded interval rises and falls with the open population over the course of the concentration jump, and it also depends on the channel number, so the constant-variance premise is wrong by construction in every cell we simulate. For LSE the quantity $\mathrm{Cov}(S)$ read against the reported information therefore measures the size of that break. It is not an information-matrix-equality failure of the same kind as a gating-aware likelihood's, where the predictive variance does follow the gating and the discrepancy is attributable to how the gating is approximated within the interval and across intervals. The Results quote LSE's ratio with that reading, and they do not read an LSE ratio and a gating-aware ratio as two values of one comparable quantity.

The same matrix repairs what it measures. The sandwich covariance

$$\Sigma = \mathbf{H}^{-1}\,\mathbf{J}\,\mathbf{H}^{-1} = \mathbf{H}^{-1/2}\,\mathbf{C}\,\mathbf{H}^{-1/2}$$

is the corrected parameter covariance: it substitutes the information the likelihood actually carries for the information it reported. This step, and the bias vector b above, are first-order approximations, from a Taylor expansion of the score at the optimum with the empirical Hessian replaced by its expectation $-\mathbf{H}$ (see Supplementary Information). We verify the correction directly, without leaning on the expansion, by comparing the sandwich-corrected ellipse against the empirical covariance of the maximum-likelihood estimates (MLEs) over the simulated recordings (Figure **??**). A diagnostic that also corrects the error bars it flags is a stronger result than one that only scores algorithms.

The distortion splits into two physically distinct parts (Figure **??**–figure supplement 3). The sample distortion $\mathbf{C}_s = \mathbf{H}^{-1/2}\,\mathbf{J}_s\,\mathbf{H}^{-1/2}$, built from the within-interval score covariance $\mathbf{J}_s = \sum_t \mathrm{Cov}(s_t)$ with $s_t = \nabla_\theta \ell_t$ the score of interval t, measures per-sample departure from Gaussianity; to the leading corrections it is the third and fourth cumulants of the interval current, the non-Gaussian shape the Gaussian predictive cannot carry. The correlation distortion $\mathbf{R} = \mathbf{J}_s^{-1/2}\,\mathbf{J}\,\mathbf{J}_s^{-1/2}$ measures what the within-interval covariance misses, the cross-interval correlation of the score, which is exactly the local time correlation of the current that **?** named as the dominant error of non-recursive fitting; $\mathbf{R} = \mathbf{I}$ when the score carries no cross-interval dependence. These compose exactly. The exact identity is

$$\mathbf{C} = \mathbf{K}\,\mathbf{R}\,\mathbf{K}^{\top}, \qquad \mathbf{K} = \mathbf{H}^{-1/2}\,\mathbf{J}_s^{1/2},$$

which holds as algebra for any $\mathbf{H}$ and $\mathbf{J}_s$, since substituting the definitions collapses the chain. The factor $\mathbf{K}$ satisfies $\mathbf{K}\mathbf{K}^{\top} = \mathbf{C}_s$, so it is a legitimate square-root factor of the sample distortion, but it is not the symmetric principal square root, and the form $\mathbf{C} = \mathbf{C}_s^{1/2}\,\mathbf{R}\,\mathbf{C}_s^{1/2}$ holds only where $\mathbf{H}$ and $\mathbf{J}_s$ commute. What composes with no assumption at all is the information volume,

$$\log\det\mathbf{C} = \log\det\mathbf{C}_s + \log\det\mathbf{R},$$

an exact identity because the determinant is multiplicative. The per-parameter diagonal, the trace, the eigenvalues and the Frobenius norm do not compose across the split; only the volume does, and only the volume is decomposed in the maps.

The correlation distortion has a reader-facing translation. For any additive statistic X_t with sum $L = \sum_t X_t$, the ratio $\kappa = \mathrm{Var}(L)\big/\sum_t \mathrm{Var}(X_t)$ is unity when the terms are independent and larger when they are positively correlated, and it converts to an effective sample size $T_{\mathrm{eff}} = T/\kappa$: a non-recursive likelihood believes it has T independent intervals when it effectively has T/κ. This is the number from the diagnostic most worth quoting, and it is exact by definition. When $\mathbf{R} \approx \kappa\mathbf{I}$ the temporal error is isotropic and this scalar captures it; when $\mathbf{R}$ is anisotropic, or when $\mathbf{C}_s \neq \mathbf{I}$, the distortion cannot be reduced to a sample-size correction and the full matrix is needed.

Two conventions govern the edge cases. The direction is fixed as above, $C_{ii} > 1$ being over-confidence and $C_{ii} < 1$ conservatism, stated once so every verdict reads the same way. And the anchor can be near-singular where a parameter is unidentifiable, which by design happens once the open population relaxes (Figure **??**–figure supplement 1); there we work in the retained eigenspace of $\mathbf{H}$ above a numerical cutoff and leave the quantity undefined, not finite, when that eigenspace is empty. Undefined is not zero, and the maps render those cells in grey to keep the two apart.

Results

The figures are ordered so that each one answers the objection the one before it raises. Figure **??** shows that the members of the cost ladder do different arithmetic on the same recording, which invites the question of whether that arithmetic changes what an experimenter takes away. Figure **??** answers it at one design cell, where the members recover the parameters and report intervals that differ by more than an order of magnitude. That raises the question of whether the fault lies in the likelihood or in the fit, so Figure **??** puts the question to the likelihood itself, scoring one common set of recordings interval by interval, and locates the failure in how the information accumulates over time. One cell is not a result, so Figure **??** repeats both moments over the plane of channel number against instrumental noise, with the acquisition interval resolved inside each cell. Given a member whose reported interval is honest over most of that plane, Figure **??** asks what a channel budget should be spent on. Figure **??** collects the answers into a map of what a macroscopic recording can be asked for.

Throughout, N_{ch} is the number of channels in the patch, k_{on} and k_{off} the opening and closing rates, i the unitary current, Δ the acquisition interval and $\tau = 1/k_{\mathrm{off}}$ the closing time constant, so that $\Delta \cdot k_{\mathrm{off}}$ is the interval in units of τ. The instrumental noise is reported as the label the runs and the figure axes carry; the dimensionless noise $\tilde{S}$, the instrumental noise power accumulated over one channel time constant measured against the unitary current, is one tenth of that label (Methods).

One filter step along the cost ladder

Figure **??** puts one turn of the filter cycle for the four body members side by side, on a single simulated recording of a patch of $N_{\mathrm{ch}} = 20$ channels. It fixes the vocabulary and carries no measured claim.

Least squares (LSE) fits the deterministic mean current and assigns one constant variance $\hat{\sigma}^2$ to every residual, estimated as the residual sum of squares over the record. Its predicted band therefore has the same width at every interval, whether the channels are gating or not, which is the homoscedastic premise of the classical fit made visible on the trace. The non-recursive member (NR) replaces that single constant with the per-interval gating variance the model predicts, so its band follows the open population, and its update row reads as an open loop: the occupancy carried into each interval comes from the initial condition and never from the data. The recursive member (R) adds the Bayesian update, on a conductance taken at a single instant. The boundary-conditioned member (IR) conditions the interval-averaged conductance on the channel states at both ends of the interval, which lets the covariance between them enter the gain. Reading left to right, each column adds one operation to the column before it, and the cost of a likelihood evaluation rises with it.

Recovery at one design cell

Figure **??** turns the ladder into a statement about reported uncertainty at one cell: $N_{\text{ch}} = 100$, noise label 0.1 ($\tilde{S} = 0.01$), $\Delta{\cdot}k_{\text{off}} = 0.1$. Ten thousand independent recordings are partitioned into groups of ten, each group is fitted by maximum likelihood under each member, and the scatter of the resulting thousand estimates is compared against the covariance that member reports from its own Gaussian Fisher information.

The two objects disagree, and by an amount that falls monotonically down the cost ladder. Taking the ratio of the area of the empirical 95% ellipse to the area of the reported one, and averaging geometrically over the six parameter pairs, the open-loop member reads 11.41 (per-pair range 10.04 to 14.83), the recursive instantaneous member 1.17 (1.08 to 1.32), and the boundary-conditioned member 1.01 (0.98 to 1.05). On the joint four-parameter ellipse the same three read 154.65, 1.56 and 1.12. Per parameter, the diagonal of the distortion matrix $\mathbf{C}$ at this cell runs $13.24/17.32/10.04/9.91$ for NR, $1.39/1.51/1.16/1.16$ for R and $1.05/1.13/1.00/0.90$ for IR, on k_{on}, k_{off}, i and N_{ch} in that order. A likelihood that carries no recursion therefore under-reports the width of the parameter distribution by a factor of ten to fifteen, while the recursive members are wrong by tens of percent. The one-endpoint member sits between the two recursive ones, at 1.63 on the same measure, and the supplement uses it to take the step from R to IR in its two algebraic halves.

The distortion is measurable and it is also removable. Replacing each reported covariance by the sandwich, which substitutes the observed covariance of the score for the information the likelihood claimed, returns every member of the ladder to within about a tenth of one: 1.03 for NR, 0.95 for R, 1.05 for IR, with every parameter pair falling between 0.87 and 1.07; the one-endpoint member carried in the supplement reads 0.90. The reader-facing form of that repair is coverage rather than area. Reducing the full multivariate distribution of the estimates to the squared Mahalanobis distance from the cloud mean, which is distributed as χ^2 with as many degrees of freedom as free parameters if the reported covariance is right, the nominal 95% region of IR covers 0.914 of the estimates under its own Fisher covariance and 0.947 to 0.949 under the sandwich, measured at $N_{\text{ch}} = 10$ and noise label 0.05 over a thousand fits in six parameters at group size 100 (Figure **??**–figure supplement 4). That a single correction repairs the whole ladder is what makes the distortion a usable quantity rather than a complaint.

The least-squares column of this figure is pending. The grid configuration of the least-squares arm holds the unitary current fixed (Methods), so it has no ellipse in the amplitude pair at all, and no calibration ellipse has been computed for it in any configuration. Its position on the same ladder is measured instead in Figures **??** and **??**, where the quantities do not require a cloud.

The calibration cascade in time

The miscalibration of Figure **??** is a property of the likelihood and not of the optimiser, and Figure **??** shows where in the recording it is produced. One common set of a thousand recordings at the same cell ($N_{\text{ch}} = 100$, noise label 0.1, $\Delta{\cdot}k_{\text{off}} = 0.1$) is scored interval by interval by each member, with

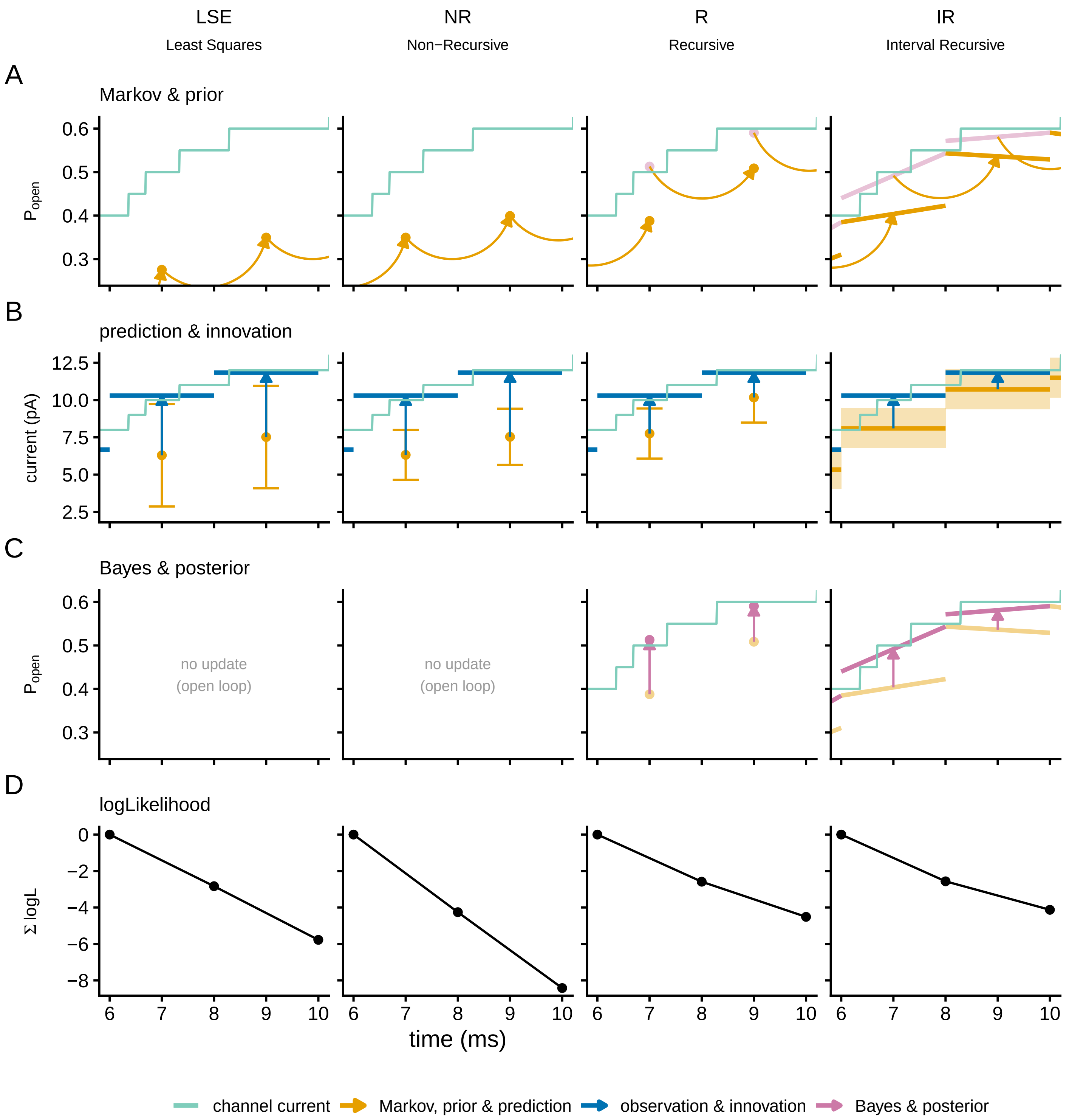


Figure 1. One filter step along the cost ladder. The current from a patch of $N_{\mathrm{ch}} = 20$ two-state channels, recorded as an interval average with white instrumental noise, is scored by each of the four body members (columns): least squares on the deterministic mean current (`LSE`), the open-loop gating likelihood (`NR`), the recursive filter on instantaneous samples (`R`), and the boundary-conditioned filter (`IR`). Rows follow one turn of the cycle: (A) the prior open probability from the Markov step, (B) the predicted current with its spread and the innovation, (C) the Bayesian posterior, which for the two non-recursive columns reads as no update, and (D) the per-step log-likelihood accumulated over the window and reset at the window start. Two measurement windows are shown (6 to 10 ms of a 12 ms recording). `LSE` is drawn in `NR`'s grammar, a point mean with an error bar, so that its constant least-squares variance can be read against `NR`'s per-interval gating variance. Colour encodes the filter phase and is shared across rows: green the simulated channel current, orange the predict phase, blue the observation and the innovation, purple the update, black the log-likelihood. A quantity carried in from the neighbouring interval is drawn hollow or dashed. The recording is one 20-channel trace simulated with 250 within-interval substeps and does not enter any quantitative comparison.

Figure 1—figure supplement 1. The same filter cycle over the first ten milliseconds of the recording, so that

Figure 2. Parameter recovery and the calibration of the reported interval, at one design cell. $N_{\mathrm{ch}} = 100$, noise label 0.1 ($\tilde{S} = 0.01$), $\Delta \cdot k_{\mathrm{off}} = 0.1$; $10{,}000$ recordings in groups of ten, so one thousand maximum-likelihood estimates per member. (A) The cloud of estimates for the kinetic pair ($\log_{10} k_{\mathrm{on}}$, $\log_{10} k_{\mathrm{off}}$) and for the amplitude pair ($\log_{10} N_{\mathrm{ch}}$, $\log_{10} i$), one column per member, with three 95% ellipses drawn at the cloud mean: the empirical covariance, the Fisher covariance the member reports, and the sandwich covariance. The first annotated number is the area of the empirical ellipse over the area of the reported one; the second is the empirical over the corrected. (B) The full four-parameter correlation structure of `IR`, whose three ellipses coincide in every pair and whose centres sit on the simulation truth. Covariances are the Gaussian-Fisher family at the pooled estimate, scaled by the reciprocal of the group size.

no fitting involved. The least-squares arm here is the six-parameter configuration, which keeps the unitary current and the noise level free so that its per-interval quantities align index for index with the macroscopic dumps (Methods); nothing measured in this subsection may be carried across to the four-parameter configuration of the later figures.

The members separate first on the log-likelihood they assign to the same data. The ensemble-mean total is -262 ± 0.73 for `LSE`, -230 ± 0.83 for `NR`, -147 ± 0.20 for `R` and -143 ± 0.24 for `IR`, a spread of 118.9 nats from the bottom of the ladder to the top. The ordering follows the ladder, and most of the span is bought by modelling the gating fluctuations at all.

The hinge of the argument is that per interval every member reports its information correctly. The variance of the per-interval score matches the per-interval Fisher information for all four, and the median of $\log_{10}(J_t/F_t)$ over the informative steps is $+0.184$ for `LSE`, -0.038 for `NR`, -0.181 for `R` and $+0.006$ for `IR` on k_{on}, with -0.145 and $+0.033$ for `R` and `IR` on k_{off}. Accumulated over the recording the same ratio reads 14.96 (95% interval 13.86 to 15.93) for `LSE`, 13.63 (12.52 to 14.95) for `NR`, 1.07 (0.975 to 1.17) for `R` and 1.08 (0.995 to 1.17) for `IR`. The two non-recursive members are within a few percent of the identity at each interval and off by more than an order of magnitude once the intervals are summed. The open-loop member also changes the sign of the discrepancy between the per-step and the accumulated statistic, and a per-sample modelling error cannot do that. Least squares does not change sign, sitting above the calibrated value at both levels, which is the signature of its constant-variance premise rather than of accumulation.

The least-squares ratio needs the qualification the Diagnostics section attaches to it. The Gaussian Fisher information of a least-squares fit presumes one constant residual variance, and the exact simulator violates that premise by construction, because the variance of a recorded interval rises and falls with the open population. The factor of fifteen is the size of that break, and it is not an information-matrix-equality failure of the same kind as the open-loop gating likelihood's, where the predictive variance does follow the gating. The two ratios are reported side by side and are not read as two values of one comparable quantity.

The mechanism is visible in the same figure. At lag one the autocorrelation of the per-interval score on k_{off} is -0.0043 ± 0.0043 for `IR`, indistinguishable from zero, against 0.191 ± 0.004 for `R`, 0.864 ± 0.0019 for `NR` and 0.856 ± 0.0022 for `LSE`. A positively correlated score makes the variance of its sum exceed the sum of the per-interval informations, which is exactly how the per-interval identity can hold while the accumulated one fails. The temporal correlation the two non-recursive members leave in their score is the local time correlation of the current that **?** named as the dominant error of independent-interval fitting, measured here at the level of the inference rather than the data.

Resolving the per-step information in time (Figure **??**–figure supplement 1) yields a second result that belongs to the measurement rather than to any member. The information about the channel number, the opening rate and the unitary current falls to zero the moment the agonist is removed and the open population relaxes: having conditioned on everything up to that point, the filter gains nothing further from watching the channels decay. The closing rate is the exception and stays informative through the relaxation, which its own kinetics govern. The shape holds for every member, so it is a property of the macroscopic observable, and it answers directly the question of when each parameter stops being measured that a recent treatment of this class of models leaves open **??**.

The design plane, in both moments

One cell is not a result. Figure **??** repeats the two moments over the plane of channel number against instrumental noise, with the seven acquisition intervals resolved inside each cell, and splits the display by moment rather than by member so that the least-squares arm stays in the same panel as the likelihoods it is being compared with. The first-moment half is evaluated at the simulation truth, where a bias is visible; the second-moment half is evaluated at the pooled optimum, where the score vanishes by construction and the information comparison is clean (Methods). The least-squares arm here is the four-parameter configuration, which holds the unitary current fixed, so it appears in the $k_{\rm off}$ rows only: with the unitary current assumed, the amplitude ridge that limits the other three members does not exist for it, and a channel-number row would read as least squares winning a contest it did not enter.

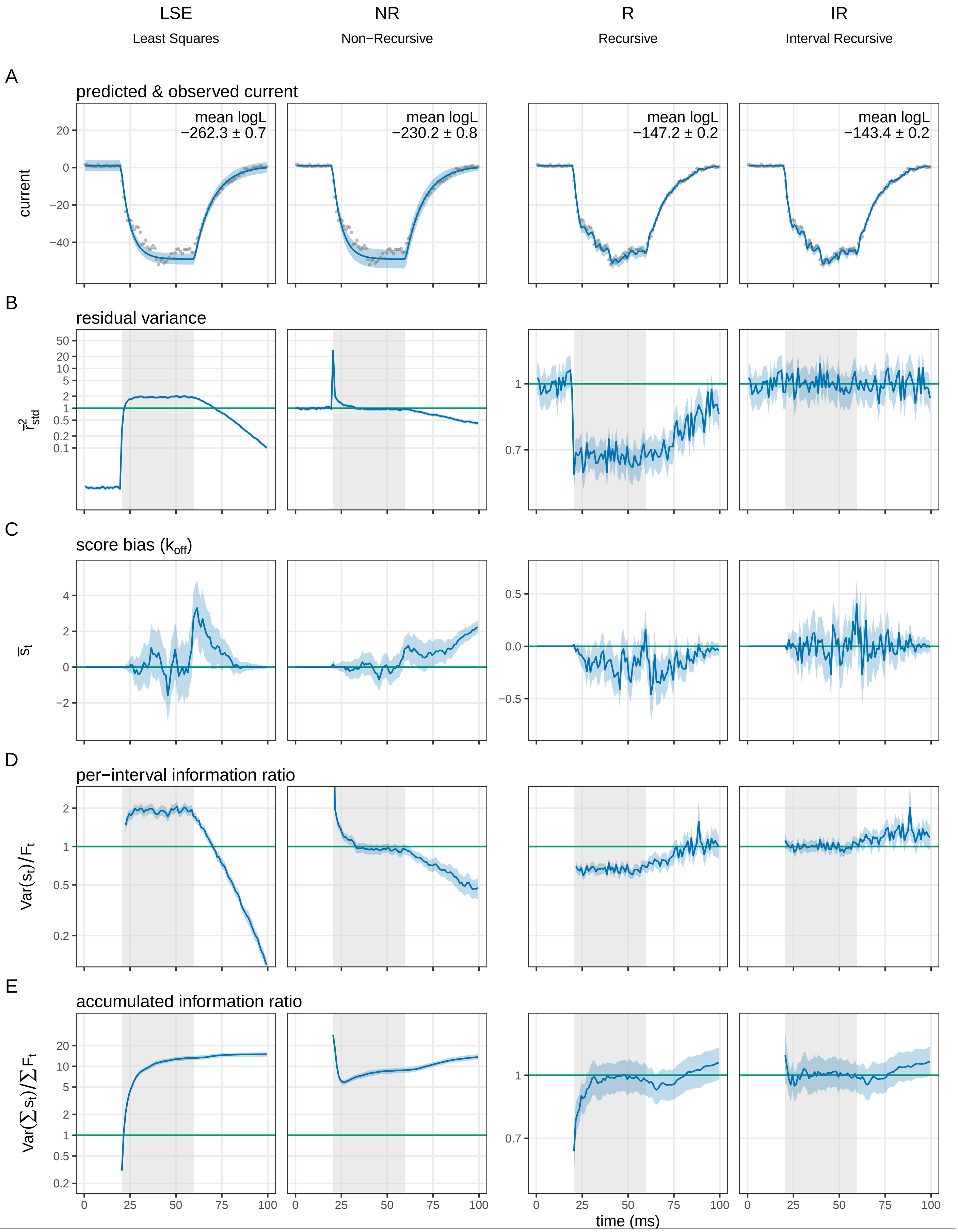
More channels do not repair the recursive instantaneous filter. Taking the median over the seven intervals at noise label 0.1, the $k_{\rm off}$ distortion of `R` sits on a flat floor of 1.372, 1.440, 1.439 and 1.42 across $N_{\rm ch}$ 10, 100, 1000 and 10^4, while `IR` converges toward one over the same range, 1.324, 1.099, 1.000, 1.00. The open-loop member moves the other way, 21.92, 16.83, 31.32 and 76.52, and the least-squares arm holds a floor an order of magnitude above one, 11.79, 13.06, 13.04 and 12.97. In the channel-number direction the two recursive members separate in sign as well as in size: `IR` runs 0.851, 0.936, 0.987, 1.00 and stays on the conservative side, while `R` crosses from conservative to over-confident, 0.866, 1.100, 1.230, 1.26, and keeps rising. The two members agree at the fewest channels and separate from there, so the 10^4 column is the informative one.

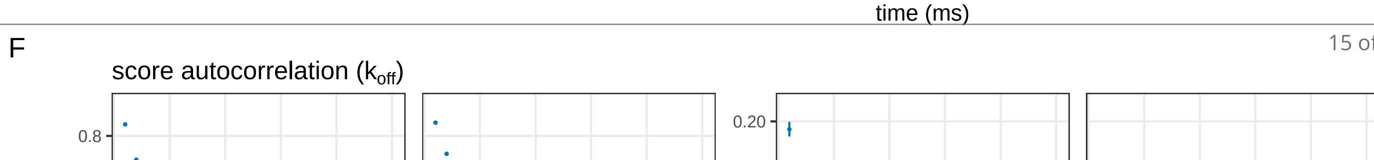
The first moment carries a separate cost. The recursive instantaneous filter has a bias in the channel number of 0.126, 0.148, 0.138 and 0.138 in $\log_{10}$ units across the same four decades, close to a 35 to 40% error that does not shrink as channels are added; the open-loop member carries 0.073 to 0.084 in the same direction, and the boundary-conditioned member is at or near zero everywhere. Least squares has a rate bias of essentially zero over the same plane. That concession is worth making plainly: the classical fit recovers the rates, and what it gets wrong is how well it says it recovered them.

The two recursive members also differ in where on the plane their departures sit, and that matters as much as their size. Counting the fraction of cells that depart from one by at least 15%, parameter by parameter, `R` departs on 0.4375, 0.5625, 0.2232, 0.3125 and 0.0000 of the plane for $k_{\rm on}$, $k_{\rm off}$, i, $N_{\rm ch}$ and the noise level, against 0.0179, 0.0804, 0.0000, 0.0268 and 0.0000 for `IR`. The distortion of the instantaneous filter is spread across parameters and across the plane; the distortion of the boundary-conditioned filter is confined to the few-channel corner. Conditioning localises the error as well as shrinking it.

Instrumental noise moves the whole picture continuously, and it moves the window-ignoring members furthest. At $N_{\rm ch} = 10$ the $k_{\rm off}$ distortion of least squares runs 11.79 at noise label 0.1, 1.14 at label 100 and 1.00 at label 1000; at $N_{\rm ch} = 10^4$ the same sequence runs 12.97, then 2.54 at label 10^4, 1.15 at 10^5 and 1.00 at 10^6. The recursive instantaneous filter follows the same pattern about three decades lower, running 1.42, 1.03 at label 100 and 1.00 at label 1000 at $N_{\rm ch} = 10^4$. The reason is that instrumental noise adds a variance that carries no gating memory, so as it grows it dilutes the temporal structure a window-ignoring likelihood fails to model, until there is nothing left to fail on. The channel count at which each member's error bar becomes honest therefore rises with the noise level, which is the crossover Figure **??** draws as a boundary.

Over the whole measured grid, counting every parameter at every cell, the reported uncertainty is within $\pm 15\%$ of the delivered one for 94% of the boundary-conditioned member's points (1977 of 2100), 70% of the recursive instantaneous member's (625 of 896), and 33% of the open-loop member's (277 of 840); the one-endpoint member carried in the supplement reaches 20%. Least squares carries no entry in this comparison, because its distortion matrix has four dimensions against the likelihood members' six and the two counts are not commensurable (Methods). The coverage


LSE
Least Squares
NR
Non−Recursive
R
Recursive
IR
Interval Recursive
A
predicted & observed current
mean logL
−262.3 ± 0.7
mean logL
−230.2 ± 0.8
mean logL
−147.2 ± 0.2
mean logL
−143.4 ± 0.2
current
B
residual variance
$\bar{r}^2_{std}$
C
score bias (k_{off})
$\bar{s}_t$
D
per−interval information ratio
$Var(s_t)/F_t$
E
accumulated information ratio
$Var(\sum s_t)/\sum F_t$
time (ms)


F
score autocorrelation (k_{off})

is honestly unequal and has to be quoted with the number: the boundary-conditioned member spans eleven channel counts from 5 to 10^4 across eleven noise levels, not on a full rectangle, and the others cover less.

The boundary-conditioned member is itself an approximation and fails in both directions. Over the whole measured grid its distortion runs from 0.645, at $N_{\mathrm{ch}} = 10$, noise label 0.05, $\Delta \cdot k_{\mathrm{off}} = 0.01$, on the channel-number direction, where it over-states its own uncertainty, to 1.701, at $N_{\mathrm{ch}} = 5$, noise label 0.1, $\Delta \cdot k_{\mathrm{off}} = 1$, on the closing rate, where it under-states it. Both excursions are in the few-channel corner, which is where the occupancy closure of the Theory section is expected to degrade.

The mechanism behind the flat floor is the split of the distortion into a per-sample part and a correlation part (Figure **??**–figure supplement 3). On the closing rate at noise label 0.1, the per-sample part goes to one as channels are added for both recursive members, 1.09 to 0.998 for `R` and 1.10 to 1.007 for `IR` across the four decades, so the single-interval Gaussian approximation becomes exact for both at the same rate. What separates them is the correlation part, which rises for `R` from 1.35 to 1.47 and falls for `IR` from 1.24 to 1.00. The two members are equally faithful to a single interval, and they part company on how much temporal dependence they leave in the score, which is what conditioning on both interval endpoints removes.

The same memory is visible in the data, without any parameter. The integrated autocorrelation of the standardized residual, which counts how many intervals a likelihood effectively has for every interval it believes it has, reads 1.008 for `IR`, 1.30 for `R`, 3.93 for `NR` and 9.26 for `LSE` at noise label 0.1. The classical fit therefore behaves as though it had about nine times more independent observations than the recording supplies, which is the degrees-of-freedom cost of fitting correlated data as though it were independent.

Two features of the plane are easy to misread and belong in the text. The first is the left edge. Where the acquisition interval is far shorter than the channel's correlation time, successive samples carry redundant information, and a faithful likelihood has to register that redundancy, so a large correlation distortion at the shortest intervals is the correct cost of oversampling rather than a defect. The second is a single cell carried openly. At $N_{\mathrm{ch}} = 10^4$ and $\Delta \cdot k_{\mathrm{off}} = 1$ the amplitude directions of the recursive instantaneous filter are not identified: the Gaussian Fisher information is ill-conditioned there, so a displacement that costs almost no likelihood is amplified into a large parameter shift, and the reported bias runs off the scale of the map. Those cells are drawn grey. Their sign and their existence are real; their magnitude is a property of the conditioning and is not a physical bias.

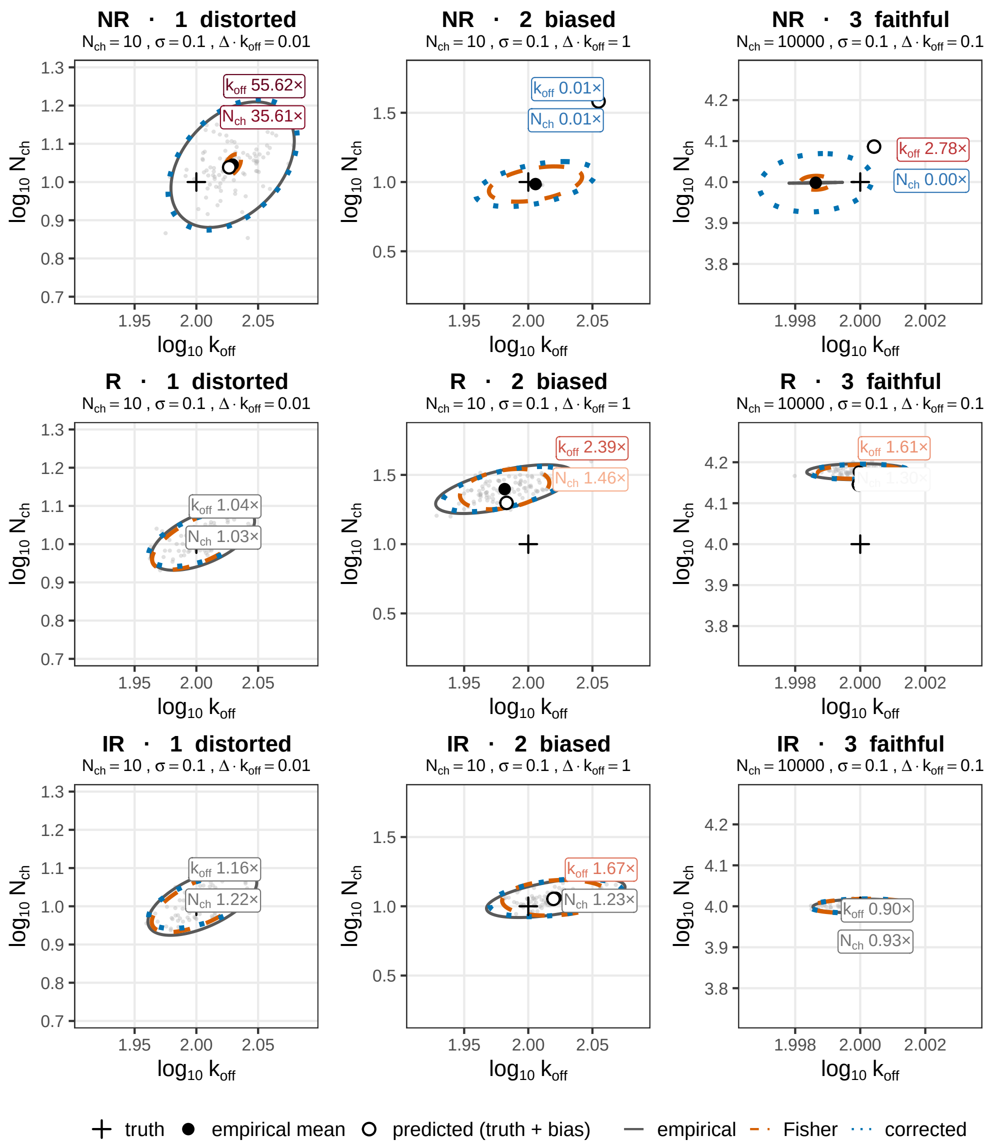
The information budget per parameter

A calibrated member's reported covariance can be spent, and Figure **??** spends it. Only the boundary-conditioned member is shown, because the quantity is a covariance the member reports and only a member whose report has been checked can guide a design. The covariance used is the distortion-corrected one, so what is plotted is the precision the experiment achieves.

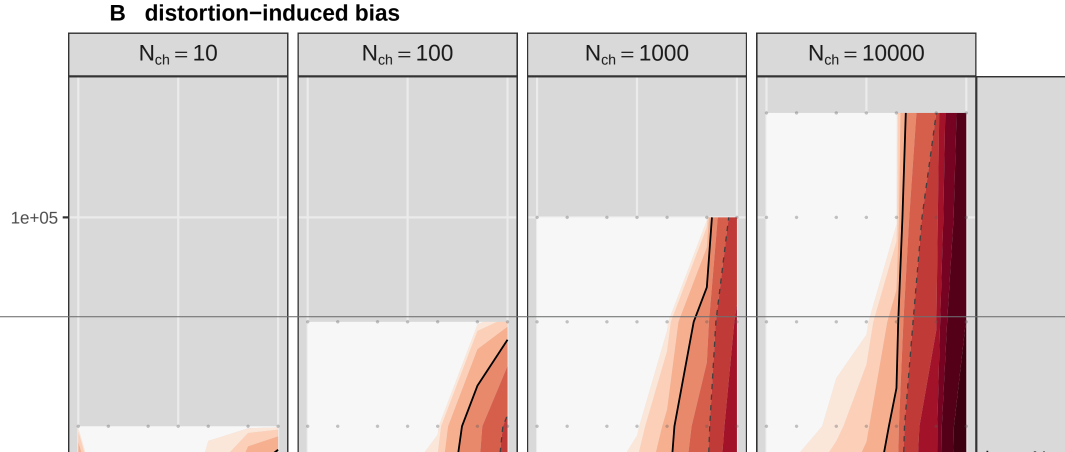
The quantity is a trade-off at a fixed channel budget. Plotting N_{ch} times the corrected variance of a parameter gives, up to the fixed budget constant, the variance of the estimate pooled over as many recordings as the budget allows. A flat line means the information is linear in channels, so it makes no difference whether the budget is spent as a few large patches or many small ones. A rising line means concentrating the budget into few large recordings yields less information.

The rates and the amplitude directions give opposite advice. Over the four decades of channel count at noise label 0.1, the pooled variance grows by a factor of 1.24 for the closing rate and 3.11 for the opening rate, against 288.19 for the unitary current, 60.50 for the channel number and 3236.85 for the instrumental noise level. For the rates, a channel budget can be concentrated at almost no cost. For the amplitude, the count and the noise level, the same budget spent as many small recordings gives a substantially better estimate. A noisier rig penalises the few-channel end, where the instrumental noise is large next to the gating variance, and it does so first in the rates, which

A grouped MLE distributions



B distortion−induced bias



are otherwise insensitive to how the budget is split. The acquisition interval runs the same way in every panel, with the pooled variance smallest at short intervals and growing toward the coarse end, mildly for the rates and more for the amplitude directions, which agrees with the calibration trade-off of Figure **??**.

The scope of this figure is narrower than the trade-off it states. Everything here is the non-stationary protocol, a single concentration jump with a rising and a relaxing phase. The rates are recoverable from the shape of that deterministic transient, which is why they cost so little. Under a stationary protocol the mean current is flat, the transient carries no rate information, and the balance between the rates and the amplitude directions inverts. The advice below should not be carried across that boundary.

A usage map for a macroscopic recording

Figure **??** collects the measurements into one plane, the channel number against the instrumental noise, at a single acquisition interval $\Delta \cdot k_{\mathrm{off}} = 0.1$. Unlike the earlier maps, its content is the boundaries: the measured cells are the input and the lines are the output. Four boundaries are drawn, and they are two questions asked of two methods. Two ask whether a method's reported error bar is honest, one for the boundary-conditioned likelihood and one for least squares, and are drawn where that method's information distortion crosses 1.15, a 7% error on the reported standard deviation. Two ask whether a parameter is reachable at all, one for the amplitude pair and one for the rates, and are drawn where the distortion-corrected standard error crosses a factor of two. The rate boundaries are taken from the least-squares arm and the amplitude boundaries from the likelihood, because the least-squares arm of this figure is the four-parameter configuration, which holds the unitary current fixed and so does not estimate it (Methods), while the two arms agree on the rates over the channel counts where both are measured and least squares is the arm with the high-noise coverage.

Read upward at a fixed channel count, the recording loses capabilities in a fixed order, which gives five regions. In region 0, below the lowest boundary, the Gaussian closure is itself misspecified, so the likelihood's own error bar is wrong and the recording is not a macroscopic problem. Region 1 delivers the unitary current, the channel number and an honest interval, while least squares reports an interval several times too narrow. In region 2 the amplitude pair is no longer separable and the rates alone survive, so the likelihood still buys an honest interval. In region 3 least squares is calibrated too and buys the same answer more cheaply. Above region 3 nothing useful is left. The boundary of region 0 is the one with a negative slope, measured at -0.56 in $\mathrm{d}\log_{10}\mathrm{noise}/\mathrm{d}\log_{10}N_{\mathrm{ch}}$, because instrumental noise makes the emission more Gaussian and so relaxes the occupancy closure while hurting everything else.

The boundary at which the classical error bar becomes honest is the one this paper predicts and then measures. In noise-label units it sits at 164 and 130 at $N_{\mathrm{ch}} = 10$, at 1.28×10^3 and 1.53×10^3 at 100, at 1.54×10^4 and 1.51×10^4 at 1000, and at 1.62×10^5 and 1.59×10^5 at 10^4, taking the channel-number and the closing-rate directions in that order; the dimensionless noise is one tenth of each. Fitted in logarithms, those crossings give slopes of 1.01 and 1.03, so the boundary runs as noise proportional to channel number. That is the crossover the mechanism predicts, since the gating variance grows with the channel count while the instrumental variance does not, so the ratio of the two is what decides whether a window-ignoring fit is honest. Here it is a measured slope rather than an assumed scaling.

The ordering of the boundaries is the result the map exists to state. There is no region of the measured plane in which the amplitude still carries information and the classical error bar is simultaneously trustworthy; the two boundaries are separated by one to three decades of noise everywhere measured. Whenever the fluctuations still say something about the unitary current or the channel number, a classical fit reports a confidence interval that is too narrow. The scope of that statement has to be stated with it: the classical arm here is least squares on the deterministic mean current with the unitary current held fixed, in its four-parameter configuration. It is not the

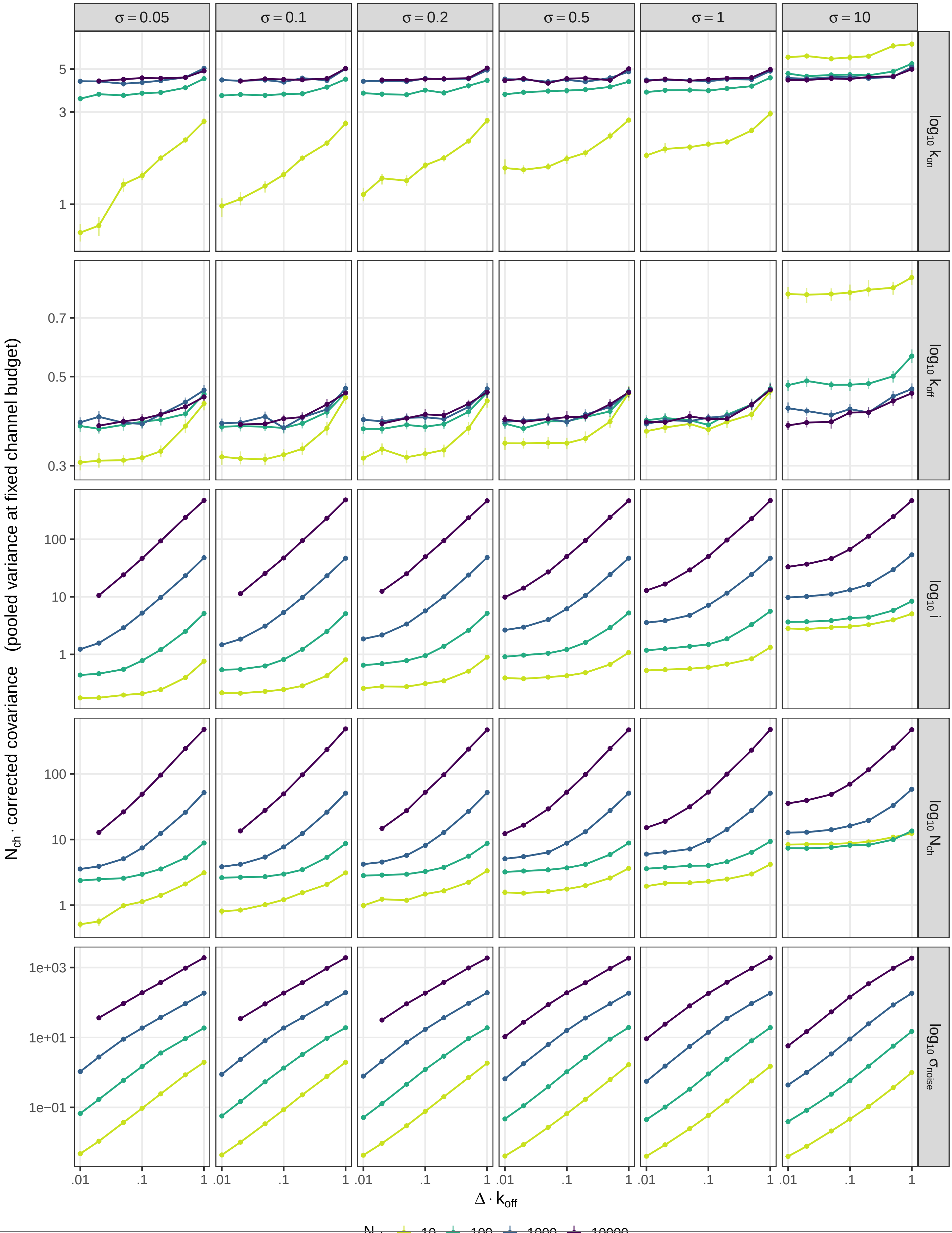


Figure 5. The information budget per parameter: how precision trades against channel count and acquisition interval at a fixed channel budget. Only the boundary-conditioned member is shown, on its

variance-mean regression, this paper does not benchmark non-stationary fluctuation analysis, and nothing here should be read as a measured comparison against it.

The map also fixes where a parameter stops being reachable at any noise level. Minimising the corrected standard error over the whole noise sweep, the channel number is unreachable below about 17 channels and the opening rate below about 52, at this acquisition interval. The unitary current and the closing rate have no such floor within the simulated range.

The evidence is thinner than the picture, and the subsection says so rather than leaving it to a caption. The two boundaries at which least squares becomes honest, and the closure floor, are each fitted through four measured crossings. The channel count at which region 3 opens spans 120 to 1368 depending only on whether the strict or the loose criterion is used. The lower vertex, where the map closes from below, is extrapolated to N_{ch} between 3 and 9, below the simulated floor of ten channels, and the ten cells that would pin it have not been run. All the boundaries are drawn at one acquisition interval, with the ribbons showing where the same boundary moves as the interval is swept from 0.01 to 1 in units of τ. The map is a concept map, not a phase diagram. Its boundaries are level sets of continuous diagnostics, so a looser criterion moves each of them by up to a decade in noise without changing the layout.

Three recording configurations are placed on the same plane, as footprints rather than points: an excised outside-out patch, a whole cell, and a two-electrode voltage-clamped oocyte. Each footprint is the range of channel numbers the configuration expresses crossed with the range of instrumental noise it carries, converted into the sweep's own axis through the unitary current and the channel time constant, and clipped to the relaxation each configuration can resolve. In the sweep's label units the patch spans about 3×10^{-5} to 2.5, the whole cell about 10^{-4} to 20, and the oocyte about 8×10^{2} to 9×10^{6}. The placement is provisional and is offered as orientation. Typical instrumental noise per configuration is not published in a form that can be used directly, so the noise ranges are read off published current traces by a calibrated procedure, and the minimum resolvable current for two of the three configurations is a working value that has not been sourced.

Discussion

The choice of macroscopic likelihood can be settled by measurement instead of by habit, and over most of the design space the measurement has one answer. Of the 2100 points at which the boundary-conditioned member was measured, one per parameter per design cell, spanning eleven channel counts from 5 to 10^{4} and eleven instrumental noise levels with seven acquisition intervals resolved inside each cell, 1977 (94%) have a reported uncertainty within 15% of the one delivered. An experimenter who does not want to consult a map can use that member everywhere and be wrong about the width of the interval it reports only in the few-channel corner. Where it is wrong, the correction that repairs it is the one the diagnostic already supplies: at $N_{\mathrm{ch}} = 10$ its nominal 95% region covers 0.914 of the estimates under the covariance it reports and 0.947 to 0.949 under the sandwich, over a thousand fits in six parameters.

That member is an approximation and is not exempt from the failure this paper is about. Over the same grid its distortion runs from 0.645, at $N_{\mathrm{ch}} = 10$ with noise label 0.05 and $\Delta{\cdot}k_{\mathrm{off}} = 0.01$ on the channel-number direction, where it over-states its own uncertainty, to 1.701, at $N_{\mathrm{ch}} = 5$ with noise label 0.1 and $\Delta{\cdot}k_{\mathrm{off}} = 1$ on the closing rate, where it under-states it. The departure is two-sided, and that is how it should be read: the reported interval moves in both directions as the channel count falls, and the approximation does not become safe by erring conservatively on average. Both excursions sit where the open population is a small integer count whose distribution is visibly discrete, which is where the occupancy Gaussian, the family's second closure, is expected to degrade. The channel count below which that closure is itself misspecified is [THRESHOLD-DJ: N$_c h, channels-or-openingspending$], $anditmoveswiththeinstrumentalnoise, downwardasthenoiserises, becauseinstrumentalvariancemake$ $withinthemacroscopic family the occupancy closure and the interval closure are applied together at every step, and the scorese$

A second boundary of the family's validity is argued here from its premise and has not been measured. Every member above least squares assumes that the instrumental contribution to a

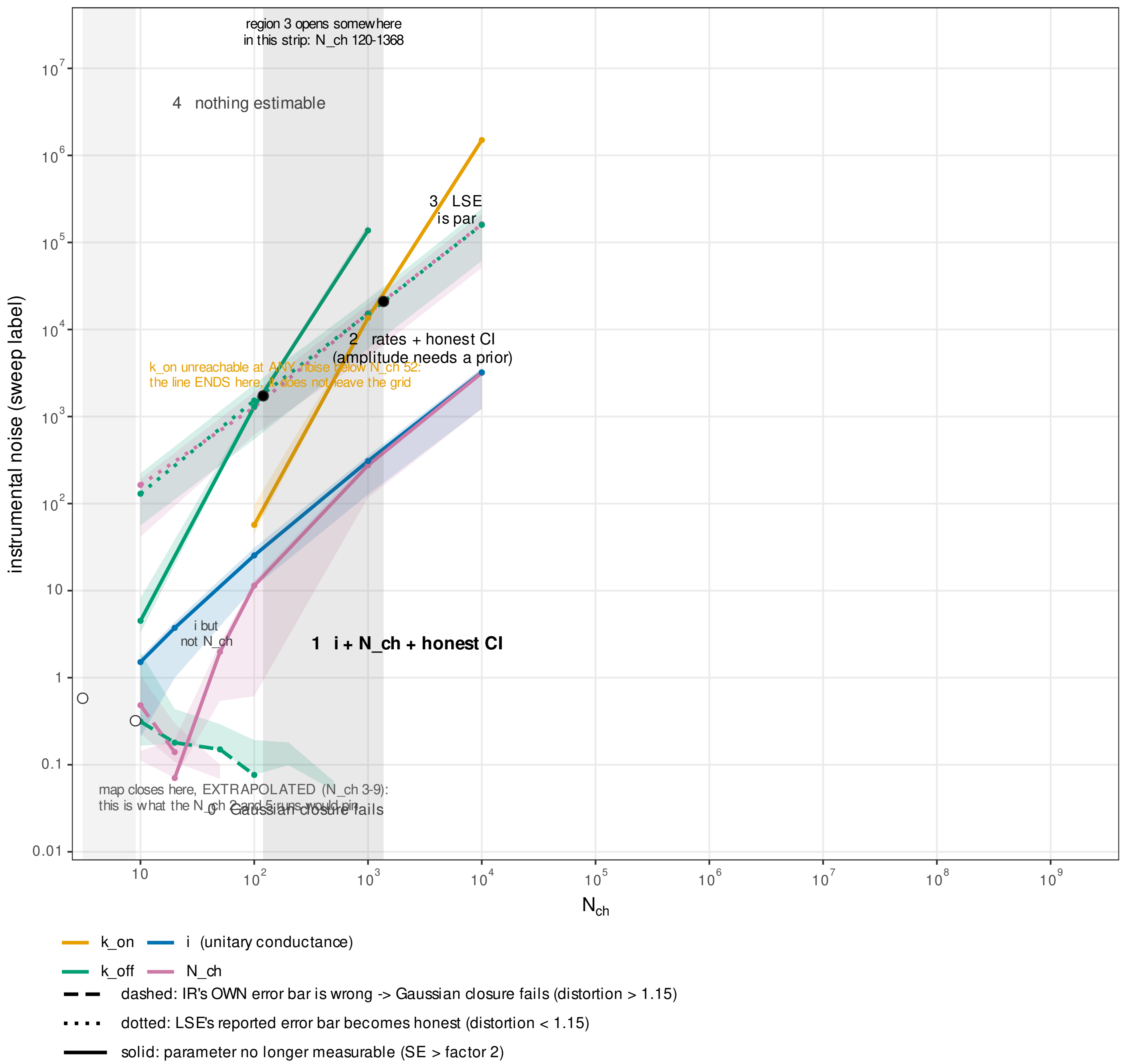


Figure 6. What a macroscopic recording can be asked for: a usage map over channel number and instrumental noise. The plane is the design space, the number of channels in the patch against the instrumental noise label, both on logarithmic scales, at one acquisition interval $\Delta \cdot k_{\text{off}} = 0.1$. Four boundaries divide it, and they are two questions asked of two methods. Colour is the parameter the boundary is about. Line type is the question: solid asks whether the parameter can be measured at all, drawn where the distortion-corrected standard error crosses a factor of two; dotted asks whether the reported error bar of least squares is honest, drawn where its information distortion crosses 1.15; dashed asks the same of the likelihood's own error bar, and is the boundary below which the Gaussian closure is misspecified. Ribbons are the same boundary swept over acquisition intervals from 0.01 to 1 in units of τ, so a ribbon says where a reader's own sampling rate puts the boundary and is not an error bar. Dots are the measured crossings and the line through them is their fitted power law, drawn faint where it leaves the simulated range, which is marked. Vertical strips mark the two vertices, each a range set by the strict and the loose criterion. The opening rate binds before the closing rate because a single concentration jump determines it only through the difference between the two relaxation rates. The map is a concept map, not a phase diagram. Its boundaries are level sets of continuous diagnostics, so a looser criterion moves each of them by up to a decade in noise without changing the layout.

Figure 6—figure supplement 1. The same plane with the five regions shaded and named, and the three recording configurations drawn as footprints, which are provisional orientation rather than measurement, since the noise ranges are read off published traces and the minimum resolvable current for two of the three is an unsourced working value; the closure floor, the two regions in which the likelihood buys something least

recorded interval is Gaussian and white, entering the predictive variance as one additive constant and leaving no memory between intervals. Recordings depart from that description, through the two-level switching of a trapped charge in the headstage and through the flicker component of the amplifier, and the discrepancy literature for models of this class already argues that unmodelled structure of exactly this kind, rather than independent Gaussian error, is what the residuals of a fitted electrophysiology model carry **?**. Where such a departure is large, the diagnostic would report a distortion that belongs to the noise model rather than to the gating closure, and the score alone does not separate the two. We have not run it. The simulator emits Gaussian white instrumental noise only, no cell with any other noise process exists in the sweep, and nothing in this paper measures the size of the effect. This is an expectation from the closure's premise rather than a result, and the way to settle it is to simulate the alternative noise process and repeat the same measurement.

What buys the calibration is conditioning at both ends of the acquisition interval. Conditioning on the start alone does not buy it, and at the reference cell it costs. The one-endpoint member and the boundary-conditioned member assign the same total variance to the observed interval average, from the same state, and they book it differently: the spread of the interval mean across end states enters the one-endpoint member as observation noise, while the boundary-conditioned member carries it as a covariance between the interval current and the occupancy at the end of the window, which is the numerator of the Kalman gain (Eq. **??**). The whole distance between the two therefore lies in the update. On the ellipse-area measure at $N_{\mathrm{ch}} = 100$, noise label 0.1 and $\Delta \cdot k_{\mathrm{off}} = 0.1$, the one-endpoint member reads 1.63 against 1.01 for the boundary-conditioned one, and it is also worse than the instantaneous recursive filter at 1.17, so the intermediate rung does not interpolate between the two ends of the step it sits in.

A supplement takes that step in its two algebraic halves and confirms a prediction that was registered before the run. The variance-recursive member removes the end-state spread from the observation variance without restoring it in the gain, and the prediction was that its reported uncertainty would come out worse than the one-endpoint member's, over-confident and more so. It does. The reading carries one qualification the algebra makes obligatory: the predictive variance divides the gain, so altering the variance form also moves the update, and the two halves are not a clean separation into a variance step and a gain step. This is a supplement result and the body's claim does not rest on it.

The diagnosis is Milescu's, now measured. **?** named the local time correlation of the current as the dominant error of estimates that do not filter, and that correlation is what the distortion decomposition isolates. Interval by interval every rung reports its information correctly; the miscalibration appears in how the intervals accumulate. It is visible directly in the autocorrelation of the score, which at lag one on the closing rate is -0.0043 ± 0.0043 for `IR`, indistinguishable from zero, against 0.191 ± 0.004 for `R`, 0.864 ± 0.0019 for `NR` and 0.856 ± 0.0022 for `LSE` (six-parameter configuration, Methods). Conditioning on both interval ends leaves a white score. A likelihood member that skips an endpoint reads sequential structure as independent evidence, which is how a per-interval identity that holds everywhere can accumulate into an information ratio off by more than an order of magnitude. The accumulated ratio of least squares is not read this way: its information presumes one constant residual variance, which the simulator violates, so its factor of fifteen measures the size of that break rather than a failure of accumulation, and it sits above the calibrated value at the per-step level too (Diagnostics).

? argue against filtering on cost, since the moment integration and the correction step run between every pair of samples. The cost is real, and what it buys is now quantified in units of misreported uncertainty. Where the instrumental noise is large next to the gating variance the cheaper members recover: at $N_{\mathrm{ch}} = 10^4$ the closing-rate distortion of the instantaneous recursive filter falls from 1.42 at noise label 0.1 to 1.03 at label 100 and 1.00 at label 1000, and least squares follows the same path about three decades higher in noise. In that part of the plane the filter's expense is hard to justify on calibration alone. Toward few channels and short intervals the correlation part of the distortion grows, and the degrees-of-freedom count a non-conditioned fit uses is wrong by a factor that no compute saving offsets: at noise label 0.1 the integrated autocorrelation of the standardized residual is 9.26 for least squares and 3.93 for the open-loop member, against 1.008 for the boundary-conditioned filter, so the classical fit behaves as though it had about nine times more independent observations than the recording supplies. The first moment is a separate charge on the same trade. The instantaneous filter carries a bias in the channel number of 35 to 40% that does not shrink over four decades of channel count, and no error-bar correction touches it. The map states both halves, so a reader can locate a recording on it and see which side of the trade it sits on.

One result here no better likelihood can overturn, because it is a property of the macroscopic observable and not of any approximation. Once the agonist is removed and the open population relaxes, the information about the channel number, the opening rate and the unitary current falls to zero, while the closing rate stays informative through the decay it governs (Figure **??**–figure supplement 1). The shape holds for every rung, so it is scope rather than a finding about conditioning, and it answers a question **?** leave open. For the bench it says that the transient carries the information: extending the record after the agonist is removed adds data without adding information about N_{ch}, k_{on} or the unitary current, so an experiment that wants those parameters should spend its samples on the jump and its rise.

The ordering of the boundaries on the map is what an experimenter can act on. There is no region of the measured plane in which the fluctuations still carry information about the unitary current or the channel number and the classical error bar is simultaneously trustworthy; the two boundaries are separated by one to three decades of noise everywhere measured, and the boundary at which the classical interval becomes honest runs with fitted log-log slopes of 1.01 and 1.03, so it scales as instrumental noise proportional to channel number. Each of those boundaries is fitted through four measured crossings at one acquisition interval. The map is a concept map, not a phase diagram. Its boundaries are level sets of continuous diagnostics, so a looser criterion moves each of them by up to a decade in noise without changing the layout. That scaling was predicted from the mechanism, since the gating variance grows with the channel count while the instrumental variance does not, and it is then measured as a slope. The concession that travels with it is that least squares recovers the rates, with a rate bias at or near zero over the same plane; what it gets wrong is how well it says it recovered them. The scope of the comparison travels with it too. The classical arm of the design-plane and map results is the four-parameter configuration of Methods, least squares on the deterministic mean current with the unitary current held fixed; the time-resolved comparison uses the six-parameter configuration, and no least-squares quantity is carried across the two. This paper does not benchmark non-stationary fluctuation analysis, no variance-mean regression is run anywhere in it, and nothing above is a measured comparison against that method.

As an algorithm, MacroIR is an integrated-measurement augmented Kalman filter, a device long known in control and target tracking **?**, and its predictions coincide with that construction to about 10^{-8}. What is specific here is the realization for an exact continuous-time Markov chain of channel occupancies, its use at the populations simulated in this paper, from 5 to 10^4 channels, where the single-molecule integrated filters do not reach, and the measurement of what the cheaper members cost. The device is conceded; the continuous-time realization **?** and the validity map are the contribution, and the recursive lineage they extend is **?** and **?**.

The full distortion matrix is reported, rather than a scalar summary, because it is the input to the next question. Near the maximum the same matrix propagates into the Bayesian evidence through a volume term $\frac{1}{2}\log\det\mathbf{C}$ and an effective-sample rescaling $\alpha^{\star} = p/\operatorname{tr}\mathbf{C}$, so whether model comparison under an approximate likelihood ranks mechanisms correctly is decided by the object measured here. That correction is derived elsewhere and is stated only as motivation. It is one bridge of a programme that assembles exact simulation, an approximate likelihood and its correction into inference on real recordings **?**.

The perimeter of these claims is a model, and it is declared as one. The scheme is two states with the open probability held at 0.5; the protocol is a single concentration jump; the data are simulated, because a diagnostic that compares what a likelihood delivers against what it reports needs a known truth to mean anything; and the inference is likelihood only, with no prior and no evidence computation. The two-state scheme isolates the closure's own error with no confounding from scheme misspecification, and the fixed open probability means that a channel count and an opening count differ throughout by a factor of two that no measurement in this paper can settle. Richer schemes, the stationary regime, the few-channel boundary and experimental data are named companions rather than omissions. The machinery itself is not specific to ion channels: simulate a process exactly, test the score against the simulation, sandwich the Fisher information, read off the distortion. It applies to any approximate likelihood for a process that can be simulated exactly but whose likelihood cannot be written down, which is most of stochastic biophysics.

Materials and methods

Minimal model and simulation

We work with a two-state channel scheme, closed (C) and open (O), the smallest scheme in which the acquisition-averaging effect we study is present. The closed-to-open transition is agonist dependent with association constant k_{on} multiplying the agonist concentration; the open-to-closed transition is agonist independent with dissociation rate k_{off}. Only the open state conducts, and the code applies a sign flip so the single-channel current is inward. At the start of every recording all channels are closed, so the occupancy is $(1, 0)$.

The scheme carries six parameters, all estimated in base-ten logarithmic coordinates. The values used to generate the data reported here, read from the production run scripts rather than from the parameter tables in the repository (the tables hold stale values the figures never used), are $k_{\text{on}} = 10\ \mu\text{M}^{-1}\,\text{s}^{-1}$, $k_{\text{off}} = 100\ \text{s}^{-1}$, a unitary conductance g set to 1 (the engine parameter `unitary_current`, written i where the noise axis is defined below), and a current baseline b set to 1 (both in units of g), while the white-noise level and the mean channel number were swept (below). Current is expressed throughout in units of the unitary conductance g, which corresponds physically to a single-channel current of roughly 1 pA. The dissociation time $\tau = 1/k_{\text{off}} = 10$ ms sets the natural time scale that normalises the acquisition-interval axis. The relaxation at the agonist condition is faster, with rate $k_{\text{on}}[A] + k_{\text{off}} = 200\ \text{s}^{-1}$; the axis uses $\tau = 1/k_{\text{off}}$ by convention.

The emission model is white instrumental noise only. Both the pink (one-over-f) and the proportional noise channels are set to zero, so the term the reader may expect from flicker noise is genuinely absent. For a measurement interval t of duration Δ_t, the predicted current has mean $\bar{y}_t = N_{\text{ch}}\,\mu_t \cdot \gamma_t + b$ and variance

$$\sigma_t^2 \;=\; e_t \;+\; N_{\text{ch}}\,v_t, \qquad e_t \;=\; \text{Current\_Noise}\,\frac{f_s}{n_{\text{samp},t}} \;=\; \frac{\text{Current\_Noise}}{\Delta_t},$$

where N_{ch} is the channel number, μ_t the occupancy mean, γ_t the interval-averaged conductance, f_s the raw sampling rate, $n_{\text{samp},t}$ the number of raw samples averaged into the interval, e_t the white instrumental variance over the interval, and $N_{\text{ch}}\,v_t$ the channel-gating variance. The white variance e_t falls as the interval lengthens, because more raw samples are averaged into each recorded value. The gating variance v_t is the single-interval conductance variance, whose form differs across the algorithm family and is derived in the Theory section.

The stimulus is a single concentration jump in three segments, a pre-pulse at $0\ \mu$M, an agonist step at $10\ \mu$M, and a post-pulse at $0\ \mu$M, with raw sampling at $f_s = 50$ kHz throughout. Each recorded measurement is the mean of $n_{\text{samp},t}$ consecutive raw samples, so the averaging window is fixed at acquisition. That window is the acquisition interval the whole study is about, and its length is set by the number of raw samples entering each recorded value.

Ground truth is generated by exact simulation of the continuous-time Markov chain (CTMC). We use uniformization with $1{,}000$ substeps per measurement interval, so the interval average of the conductance is realised from an actual sampled trajectory rather than from any moment-closure approximation. This is what licenses treating the simulation as the reference against which the Gaussian likelihood approximations are judged: the approximations are the object under test, and the simulator supplies the exact answer they are compared to. Every maximum-likelihood grid cell reported here, the least-squares arm included, used 10^4 recordings; the time-resolved run used $1{,}000$ recordings; the single-recording illustration (Figure 1) simulated one 20-channel trace with 250 within-interval substeps for visualisation and does not enter the quantitative comparison.

The simulation seed was set to 0, which in this engine is the sentinel for a random seed drawn from `std::random_device`, and the resolved seed was never logged. The deposited ensembles are therefore statistically equivalent to, but not bit-for-bit reproducible from, a rerun of the same script. Because each ensemble is large ($1{,}000$ to $10{,}000$ recordings), the reported statistics are stable; the Monte-Carlo standard here is statistical equivalence, not bit identity.

The likelihood family and its members

The members compared here answer one question first, then a second. The first question is whether the gating fluctuations are modelled at all. Classical nonlinear least squares answers no: it fits the deterministic mean current and treats every departure from it as independent measurement error. The macroscopic likelihoods answer yes, and among them the second question is how much of the acquisition interval the interval-averaged conductance is conditioned on. Least squares is the bottom rung of that cost ladder and the comparison anchor of this paper.

Least squares is selected by the family flag *family approximation* $= 2$, its data key is `nonlinearsqr`, and it runs non-recursive with the averaging flag at 1; the variance flag is accepted and ignored on this branch. It sits off the lattice of endpoint and recursion flags that organises the macroscopic members, so it carries no compositional name. The macroscopic members all take *family approximation* $= 0$ and are one filter run under flags that set how the interval-averaged conductance is conditioned. The *recursive* flag sets whether the occupancy covariance is propagated between intervals. The *averaging* flag $\mathrm{av} \in \{0, 1, 2\}$ counts the number of interval endpoints the hidden state is conditioned on. A third flag, the *variance form*, distinguishes the two members that condition on the interval start: MR predicts the interval variance with the total per-start-state conductance variance, while VR removes the between-endpoint part, keeping MR's mean and gain.

Member	Data key	Recursive	av	Variance	Conditioning of the conductance
LSE	`nonlinearsqr`	no	1	n/a	none; the mean current only
NR	`macro_NR`	no	0	n/a	instantaneous; the acquisition averaging is ignor
NMR	`macro_NMR`	no	1	total	interval mean given the start state, without recu
R	`macro_R`	yes	0	n/a	instantaneous; the acquisition averaging is ignor
MR	`macro_MR`	yes	1	total	interval mean given the start state
VR	`macro_VR`	yes	1	residual	MR's mean and gain, with the between-endpoint
IR	`macro_IR`	yes	2	total	interval mean given both boundary states; the e

Table 2. The members as flags. LSE takes *family approximation* $= 2$ and every macroscopic member takes *family approximation* $= 0$. Reading the endpoint flag as a ladder gives no endpoints (NR, R), one endpoint (MR, VR), then both boundary states (IR). MR and IR predict the same total interval variance and differ only in the gain; VR shares MR's gain but removes the between-endpoint part of the variance, which isolates the variance change from the gain change. LSE and the macroscopic members are written by different producers, so their output files carry different prefixes, `figure_3_LSE_` and `figure_3_G_`. NMR is the member published as MacroINR in **?**, where it served as the control against which the boundary-conditioned member was compared; it is reported here only in the supplement, because over the range measured it is not separable from NR.

Table **??** lists the members. Reading the endpoint flag as a ladder, the macroscopic members condition the conductance on progressively more of the interval: no endpoints, one endpoint, then both boundary states, with the exact trajectory (what the simulator supplies) as the unreachable top rung. `IR` is the top implemented rung: it conditions on the pair of channel states at the two ends of the interval, which we call the boundary state, and marginalises the interior analytically. It is the likelihood introduced for the P2X2 receptor (**?**), here placed as one member of the completed grid, and it is verified in the Theory section to coincide with a time-integrated Kalman filter. The construction of the interval-averaged mean and variance, the boundary-state covariance, and the Kalman-like update are derived in the Theory section and its supplement; we do not restate them here.

Two reductions are worth stating because they anchor the family to prior work. Setting the boundary-conditioned conductance variance to zero in the recursive, no-endpoint member (R) recovers the macroscopic-fluctuation update of **?**, so R is the recursive ancestor already in print. The single structural difference between the M and I families is that the one-endpoint members drop the boundary cross-covariance term $N_{\mathrm{ch}}\,\gamma^{\mathrm{T}}\Sigma\gamma$ that IR retains.

Throughout the reported runs the remaining build flags were held fixed: *taylor variance correction* off, *micro approximation* off, and the *variance approximation* on. The V in VR names the *variance form* flag, set to 1, and has no connection to the *taylor variance correction* flag, which is off in every run reported here; the Taylor-corrected variants MRT and IRT are not part of this paper. Because the Taylor variance correction is off, the recursive one-endpoint member that runs is MacroMR and not MacroMRT.

The filter carries two numerical safeguards, present because the code runs them and because anyone reimplementing the filter meets the same constraint on day one. The occupancy mean is held on the probability simplex, and the occupancy covariance is kept positive semidefinite through a trust-region shrink of the Kalman-like mean update. Two constants enter, a binomial-count floor set to 5 and a minimum occupancy probability set to 10^{-12}.

The two least-squares configurations

The least-squares arm runs in two configurations, and they are different statistical models. Both select *family approximation* $= 2$, both fit data from the same simulator, and both minimise the same sum of squares. They differ in which parameters are free, so no quantity computed under one may be compared against the same quantity computed under the other.

The six-parameter configuration is `ops/local/figure_3_time_LSE.macroir`, and it feeds the time-resolved figure (Figure 3). All six parameters are free in base-ten logarithmic coordinates: k_{on}, k_{off}, the unitary current, `Current_Noise`, the current baseline and the mean channel number. The script's own header records why. The time-resolved figures accumulate the per-interval Fisher information and never invert it, so the flat directions of the least-squares fit are harmless there; and keeping six free preserves the parameter-index alignment with the macroscopic dumps that the figure code joins against, whereas fixing two would emit four indices against the macroscopic files' six and misalign them without any visible error. Two flat directions are then visible rather than hidden. `Current_Noise` has an identically zero score row and an identically zero Fisher row, because the least-squares mean carries no noise term; and the unitary current and the mean channel number form a rank-one degenerate block along the amplitude ridge $N_{\mathrm{ch}}\,i$, so the unitary current is free in this configuration without being separately identified.

The four-parameter configuration is `ops/local/figure_4_LSE.macroir`, which produces the grid cells, together with `ops/local/figure_3_mle_LSE.macroir`, which is the same model run with the per-group estimate cloud retained for the few cells that need one. Both fix the unitary current at 1 and `Current_Noise` at the cell value, leaving four free parameters: k_{on}, k_{off}, the current baseline and the mean channel number. This is the configuration behind every least-squares cell of Figures 4 and 6. Its parameter indices are $0 = k_{\mathrm{on}}$, $1 = k_{\mathrm{off}}$, $2 =$ current baseline, $3 =$ mean channel number. There is no unitary-current index in these files, and index 2 is the baseline.

The consequence is carried into the Results. A score, a Fisher information, a distortion matrix or a log-likelihood computed under six free parameters is a different object from the one computed under four, and the two distortion matrices differ in dimension, 6×6 against 4×4, so any scalar that accumulates over eigenvalues is not comparable between them. Every statement about least squares in the Results is scoped to one configuration and names it. The statement that least squares cannot deliver the unitary current is a statement about the four-parameter configuration.

Parameter estimation and the two anchors

Inference in each grid cell is two-stage. First, the ensemble is partitioned into groups and a separate maximum-likelihood estimate (MLE) is computed per group, producing a cloud of estimates whose scatter is the empirical parameter covariance. The optimiser is Gauss–Newton with Levenberg damping (initial damping 10, growth factor 3, maximum damping 10^{10}, at most 100 iterations, relative gradient tolerance 10^{-6}, value tolerance 10^{-10}), warm-started at the simulation truth. Two group sizes were used, 10 and 100; with $10{,}000$ recordings these give $1{,}000$ groups of 10 and 100 groups of 100. Second, a single joint MLE is computed over all replicates at once (one group of size n_{sim}), giving the pooled estimate θ_{pool}; its near-zero gradient certifies that it is a stationary point of the pooled likelihood.

Warm-starting the per-group fits at the truth is deliberate. It isolates the error of the likelihood from the error of the optimiser, which is what this paper studies; it would be the wrong choice for a study of an estimator, which this is not.

The two anchors, the simulation truth θ_{sim} and the pooled fit θ_{pool}, are distinct points because the Gaussian macroscopic likelihood is misspecified with respect to the exact simulator. Their difference $\theta_{\mathrm{pool}} - \theta_{\mathrm{sim}}$ is the misspecification bias. Every diagnostic below is computed at both anchors, and the two answer different questions: at θ_{pool} the score is zero by construction so bias washes out, whereas at θ_{sim} the bias is visible but the Fisher information can be indefinite away from the maximum. Each figure states which anchor it uses.

Diagnostics and the Gaussian-Fisher anchor

Faithfulness of each approximate likelihood to the simulator is measured with likelihood-level diagnostics: standardized residuals, the score (the gradient of the log-likelihood), and an information-distortion matrix that compares the covariance J of the score across replicates with the model's own Fisher information. The definitions, sign conventions, and thresholds of these diagnostics are owned by the Diagnostics section and its metric registry; here we describe only how they were computed.

Every diagnostic reported in this paper is anchored on the analytic Gaussian Fisher information $\bar{G} = \sum_t \mathrm{GFI}_t$, summed over intervals. This is the anchor the Diagnostics section writes $\mathbf{H}$, that is $\mathbf{H} = \bar{G} = \sum_t \mathrm{GFI}_t$. Its per-interval term is

$$\mathrm{GFI}_t = \frac{(\partial_\theta \bar{y}_t)(\partial_\theta \bar{y}_t)^\top}{\sigma_t^2} + \frac{(\partial_\theta \sigma_t^2)(\partial_\theta \sigma_t^2)^\top}{2\,\sigma_t^4},$$

which follows from the analytic derivative alone and needs no extra likelihood passes. It is positive semidefinite by construction, so no anchor used here can be indefinite.

A second Fisher construction exists in the codebase and was used in an earlier battery: a central difference of the analytic score, with per-coordinate step $h_i = h_{\mathrm{rel}} \max(|\theta_i|, 1)$ at $h_{\mathrm{rel}} = 10^{-5}$, symmetrised as $(F_{\mathrm{num}} + F_{\mathrm{num}}^\top)/2$. Because the score itself is obtained by automatic differentiation, this is a single numerical derivative rather than two. That battery (data directory `433ed13`) is retained in the deposit as the demonstration that the two constructions target the same information, an agreement the time-resolved figures exhibit per interval. No number reported in this paper is taken from it.

Comparing the score covariance J against the Fisher information is the sandwich, or information-matrix-equality, diagnostic (**?**): equality $\mathbf{H} = J$ holds only when the likelihood is correctly specified, and the extent to which the two differ is what the distortion matrix quantifies. The comparison has a further premise on the least-squares branch, where $\bar{G}$ presumes homoscedastic residuals; the Diagnostics section states that premise where the distortion is defined.

Uncertainty on every reported statistic comes from a nonparametric bootstrap over groups, reported as percentile intervals at the 2.5, 16, 50, 84 and 97.5% quantiles. Despite the internal column names, no probit transform is applied; the intervals are ordinary empirical percentiles, verified equal to the R quantile of type 6. The diagnostic battery used 100 bootstrap replicates and a maximum lag of 10 for the score autocorrelation; the empirical-versus-theoretical distortion capstone used 200 replicates; and fewer than 10 groups leaves the interval slots empty (not-a-number filled), which is one source of empty cells in the maps. The raw per-group MLE cloud is written unbootstrapped; any resampling of it is done downstream in the figure code.

Regime grid and reduction

The acquisition interval was swept over seven levels of Δ/τ, namely 1, 0.5, 0.2, 0.1, 0.05, 0.02 and 0.01, realised as 500, 250, 100, 50, 25, 10 and 5 raw samples averaged per interval, that is step durations of 10, 5, 2, 1, 0.5, 0.2 and 0.1 ms, giving 10, 20, 50, 100, 200, 500 and $1{,}000$ steps in the record. The total record duration is held fixed at 100 ms across the sweep, so the interval is varied without varying the amount of data. Within each record the pre-pulse, agonist and post-pulse segments hold a fixed ratio of $20{:}40{:}40$ percent of the steps.

The noise axis needs its conversion stated once, because the quantity the figures plot and the quantity the engine consumes differ by a fixed factor. The engine parameter is `Current_Noise`, a white-noise power spectral density in units of g^2 s, and it reaches the likelihood only through the variance of one recorded point, $e_t = \text{Current\_Noise}/\Delta_t$, with no correlation time of its own. The sweep is declared by a label, and the dispatchers pair each label with an engine value by a hand-written case statement, so that $\text{label} = 1000\,\text{Current\_Noise}$ exactly. The figure label 0.1 is $\text{Current\_Noise} = 10^{-4}$. The dimensionless noise is

$$\tilde{S} = \frac{\text{Current\_Noise}\;k_{\text{off}}}{i^2},$$

with i the unitary current, so $\tilde{S}$ is the instrumental noise accumulated over one channel time constant $\tau = 1/k_{\text{off}}$, measured against the single-channel signal power. At the run values $k_{\text{off}} = 100\ \text{s}^{-1}$ and $i = g = 1$, this gives

$$\tilde{S} = \frac{\text{label}}{10},$$

so the label is ten times the dimensionless noise it is named after. The factor of ten is the reference sampling interval $\Delta = 0.1\,\tau$ folded into the hardcoded divisor, and it is a naming artefact rather than a design choice; the simulations consumed `Current_Noise` and are unaffected by it. Every stored label is left as written, because those strings tie each figure to the cell that produced it. The figures are drawn in label units and their axes say so; wherever an absolute noise level appears in prose we quote $\tilde{S}$. The reference cell used throughout, label 0.1, is $\tilde{S} = 0.01$, and the label 10 cell is $\tilde{S} = 1$, the level at which the instrumental noise over one channel time constant equals the unitary current.

The mean channel number N_{ch} and the noise level are the other two map axes. The results reported here are read from three data directories, each named by the git commit hash of the build that wrote it: `1c2ae6f`, `87889e6` and `0ffbda7`, searched in that order with the first hit winning. Every cell they contribute to a body figure is at $n_{\text{sim}} = 10^4$, which the figure code enforces by matching `nsim_10000` in the filename, so a cell run at a smaller ensemble cannot enter a map silently. Across the three directories, `IR` covers $N_{\text{ch}} \in \{5, 10, 20, 50, 100, 200, 500, 1000, 2000, 5000, 10000\}$ and noise labels from 0.05 to 10^6; NR covers $N_{\text{ch}} \in \{10, 100, 1000, 10000\}$ and R covers those and additionally 5, 20 and 50 at label 0.1, with the highest labels run only at the larger channel counts and reaching 10^7 at $N_{\text{ch}} = 10^4$; MR and VR cover the same channel numbers at labels $\{0.1, 1, 10\}$; and the least-squares arm covers 30 cells on $N_{\text{ch}} \in \{10, 100, 1000, 10000\}$, with labels from 0.1 up to 10^4, 10^5, 10^6 and 10^7 respectively. The label range is deliberately not rectangular: the highest noise labels were run only at the larger channel numbers, where the crossover the map measures sits. The figure code auto-detects which cells exist on disk at render time and prints the detected grid into the render log, so the cell set behind each figure is recoverable from that log.

Each grid cell is reduced through six stages: the per-group MLE cloud; the pooled joint fit θ_{pool}; the Fisher information at both anchors; the score at both anchors, whose across-replicate covariance is J; the diagnostic battery paired with the Fisher at each anchor; and the empirical-versus-theoretical distortion capstone, anchored at θ_{pool}. Each cell writes five comma-separated-value families (the estimate cloud, the pooled fit, the two anchor batteries, and the capstone). On the grid-production lane the cloud stage and the capstone are dropped, because no map reads them there; the cells that need a cloud are produced by the separate script that keeps that stage.

Reproducibility and computational environment

The engine is `macro_dr`, built as C++20, and each analysis is a script in a small typed domain-specific language run in a single process (**?**). Provenance is self-documenting at three levels: on every invocation the binary snapshots the assembled script and its full argument list into a run ledger; every analysis output file stamps the build's short git commit hash as its first row; and the data directories are named by that hash, so results from different code versions can never overwrite one another and multi-commit provenance is documented per file.

The production batteries ran under SLURM at 32 CPUs per task and 48 GB of memory (96 GB for the parameter-recovery lane), with the linear-algebra back end pinned to a single thread so that OpenMP parallelises the per-simulation and bootstrap loops. The environment variable `MACRODR_AXIS_SERIA` is load-bearing: it serialises the internal grid-combination loop, without which memory grows with the number of concurrent combinations and the jobs exhaust memory.

The one reproducibility limit worth stating twice is the simulation seed. Because the seed was the random sentinel and the resolved value was not recorded, the exact deposited ensembles cannot be regenerated bit for bit; a rerun yields a statistically equivalent, numerically different ensemble. All resampling that operates on a fixed dataset (the bootstraps, the group partitioning) is deterministic and reproducible given the data. Deposition of the data and code, with pinned versions, is stated in the Data and code availability section.

Data and code availability

All code needed to reproduce the study is openly available. The analysis pipeline that generates the figures, together with the simulation configurations that define every run, is deposited in a public repository (`macroir-validity`) and archived at Zenodo, with the DOI issued on acceptance. The simulation and inference engine is the `macro_dr` software **?**, referenced at the pinned version that produced the deposited data; each output file records the engine commit hash that generated it. No experimental data were used: every dataset in the paper is simulated, and is regenerable from the deposited configurations.

A separate library, `macroir`, carries the boundary-conditioned likelihood, its score and its Gaussian Fisher information out of the engine and into a small self-contained C++ core, with an R package and a Python package over it. The scope of that library is narrower than the paper: it implements the boundary-conditioned member, the exact simulator and a Gauss-Newton fit, and it does not implement least squares, the open-loop member or the recursive member on instantaneous samples, so the comparisons reported here are reproducible only from `macro_dr`. The core was transcribed from the engine and checked against it on eight cells, covering four kinetic schemes, three acquisition intervals spanning a factor of a hundred, a protocol whose concentration jumps split sampling intervals, and a recording with a gap, agreeing on the total log-likelihood to between 10^{-16} and 6×10^{-15} relative and on the score to between 8×10^{-14} and 1.4×10^{-9}. The two language packages are bindings over that one core rather than independent reimplementations, and they were run against it once on a single cell, by `tools/cross_language_check.py`, which is the evidence for the statement that the three interfaces return the same numbers.

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Author contributions
L.M. conceived the study, developed the theory, implemented the algorithms, designed and ran the simulations, analysed the data, and wrote the manuscript.

Competing interests
The author declares no competing interests.

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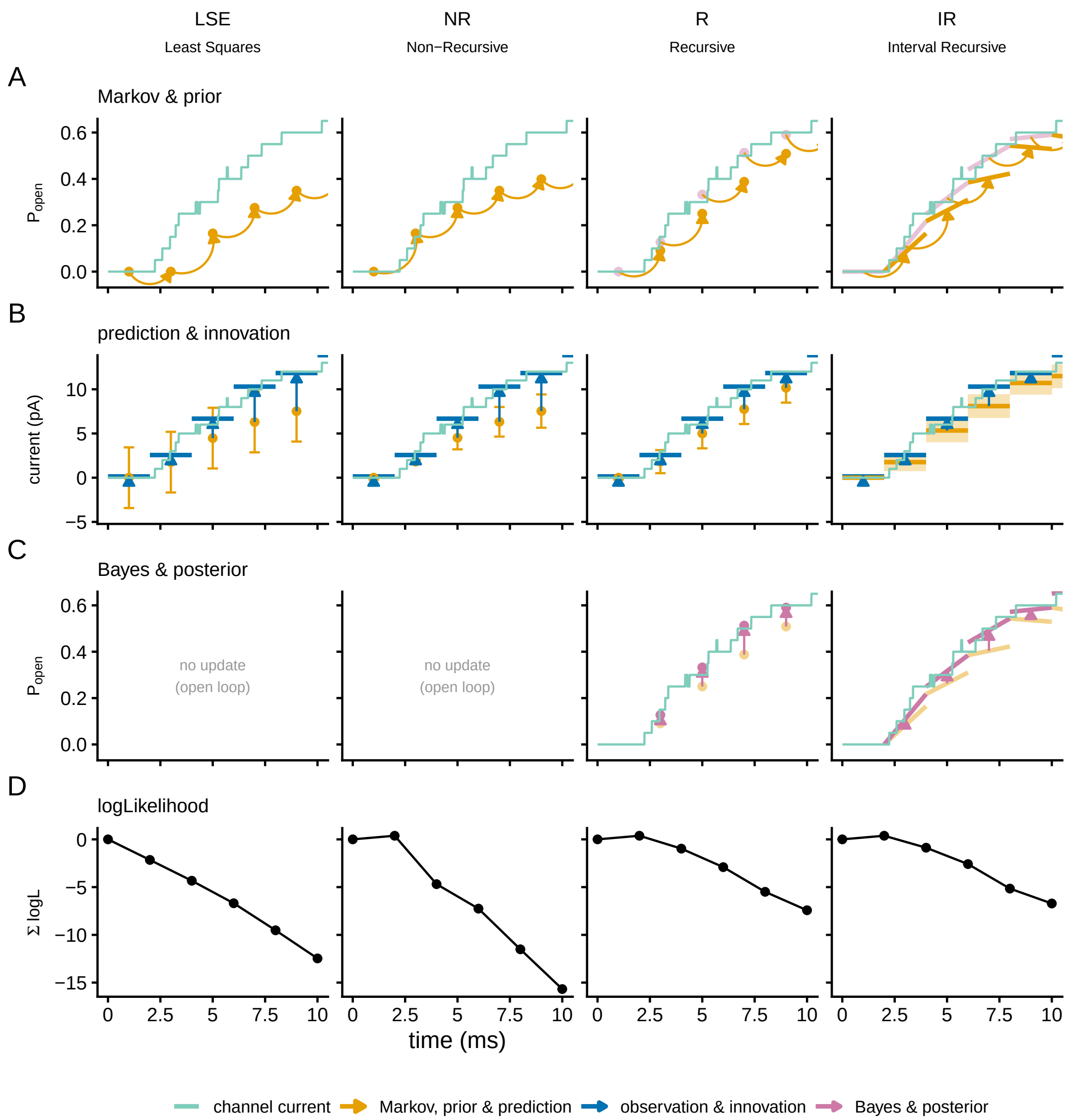


Figure 1—figure supplement 1. The same filter cycle over the first ten milliseconds of the recording, so that the concentration jump and the onset of gating are inside the window.

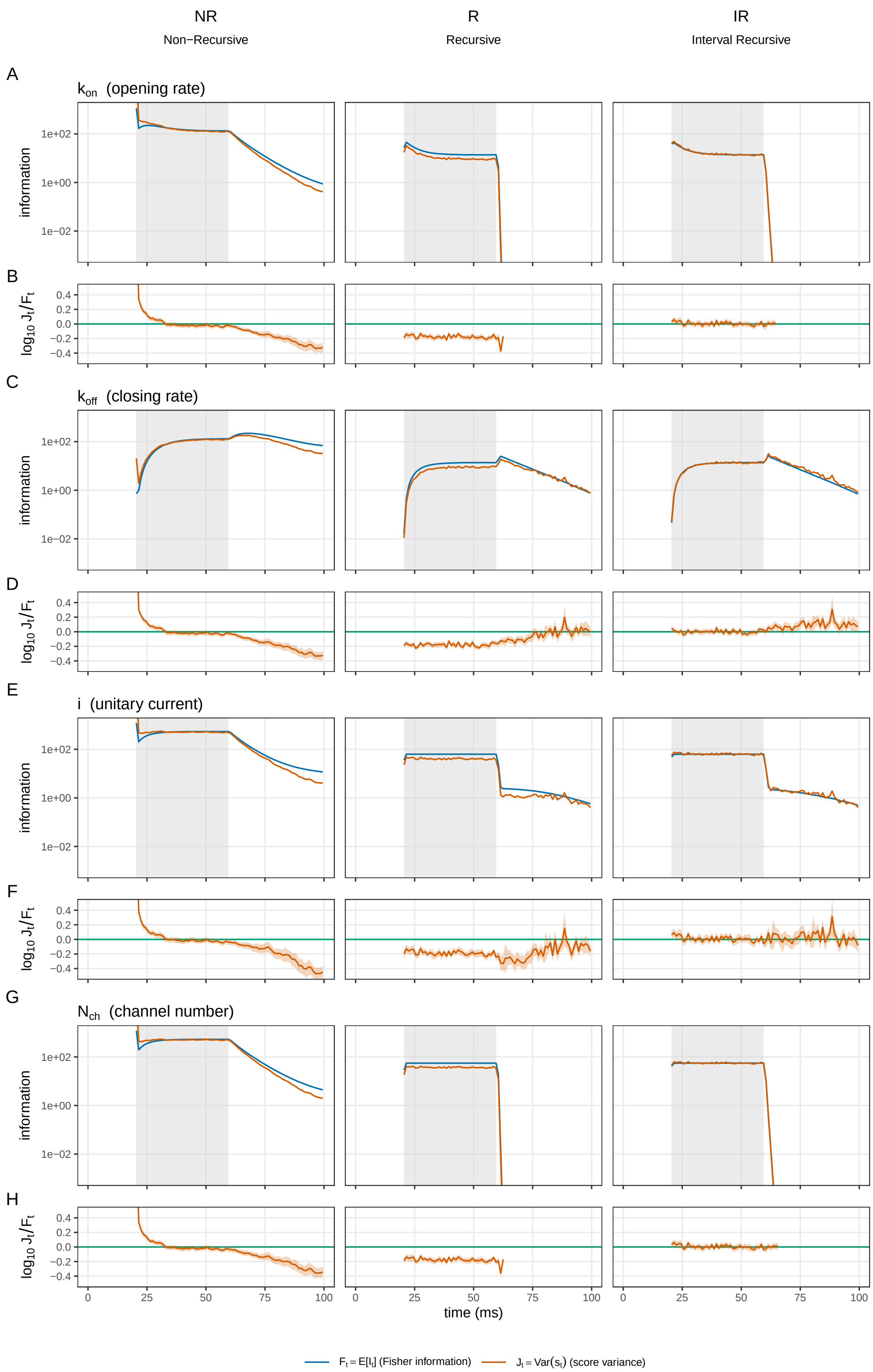


Figure 3—figure supplement 1. Where in time each parameter is measured, and the per-step calibration of that information, across the wider implemented roster and all four parameters. The information about N_{ch}, k_{on} and i falls to zero when the agonist is removed, while k_{off} stays informative through the relaxation, and the shape is the same for every member.

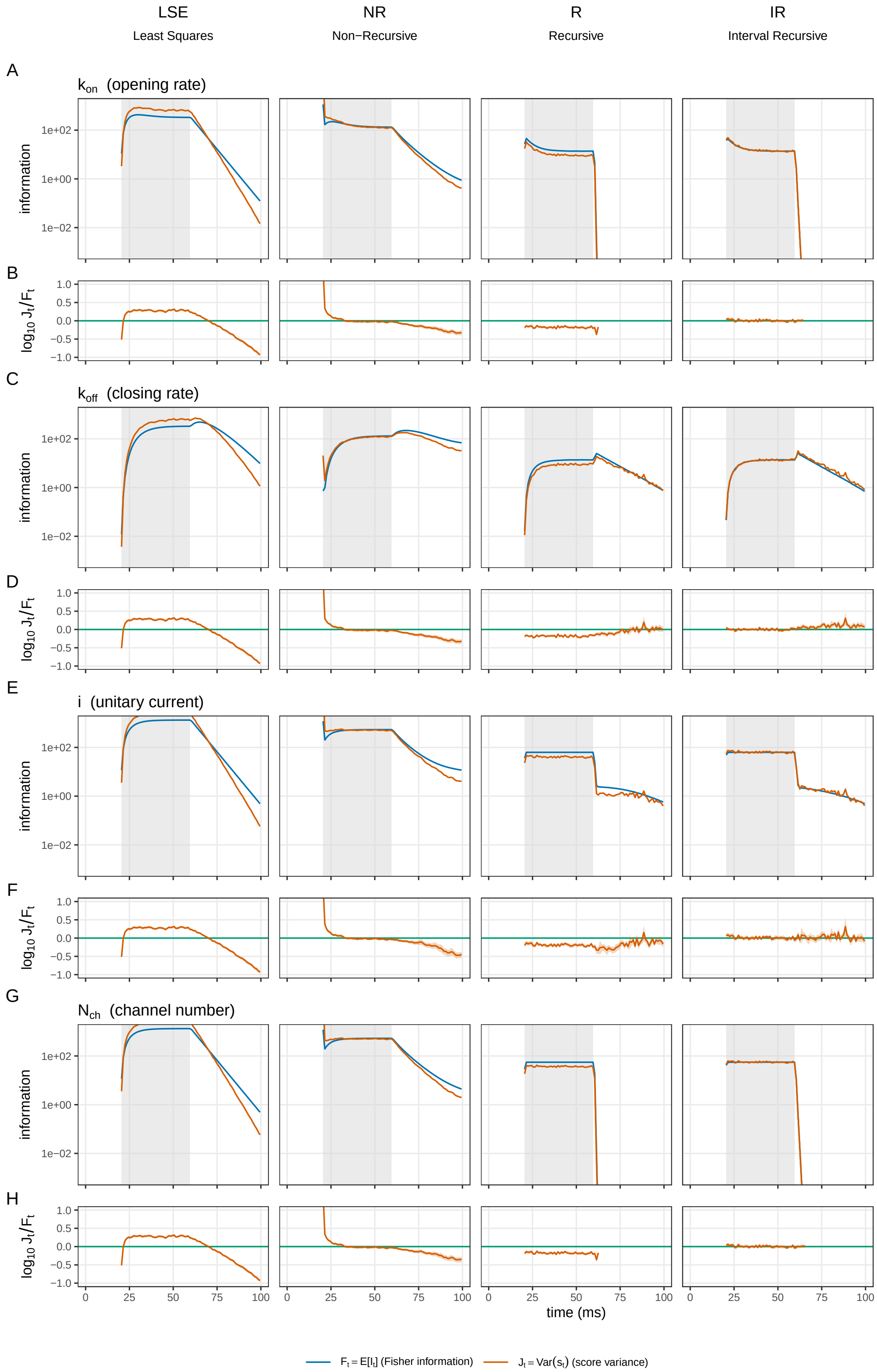


Figure 3—figure supplement 2. The same per-step comparison for the body roster, least squares against the likelihood family. The three gating-aware members sit on the calibrated value at every informative step; least squares sits above it, which is the per-step signature of its constant-variance premise.

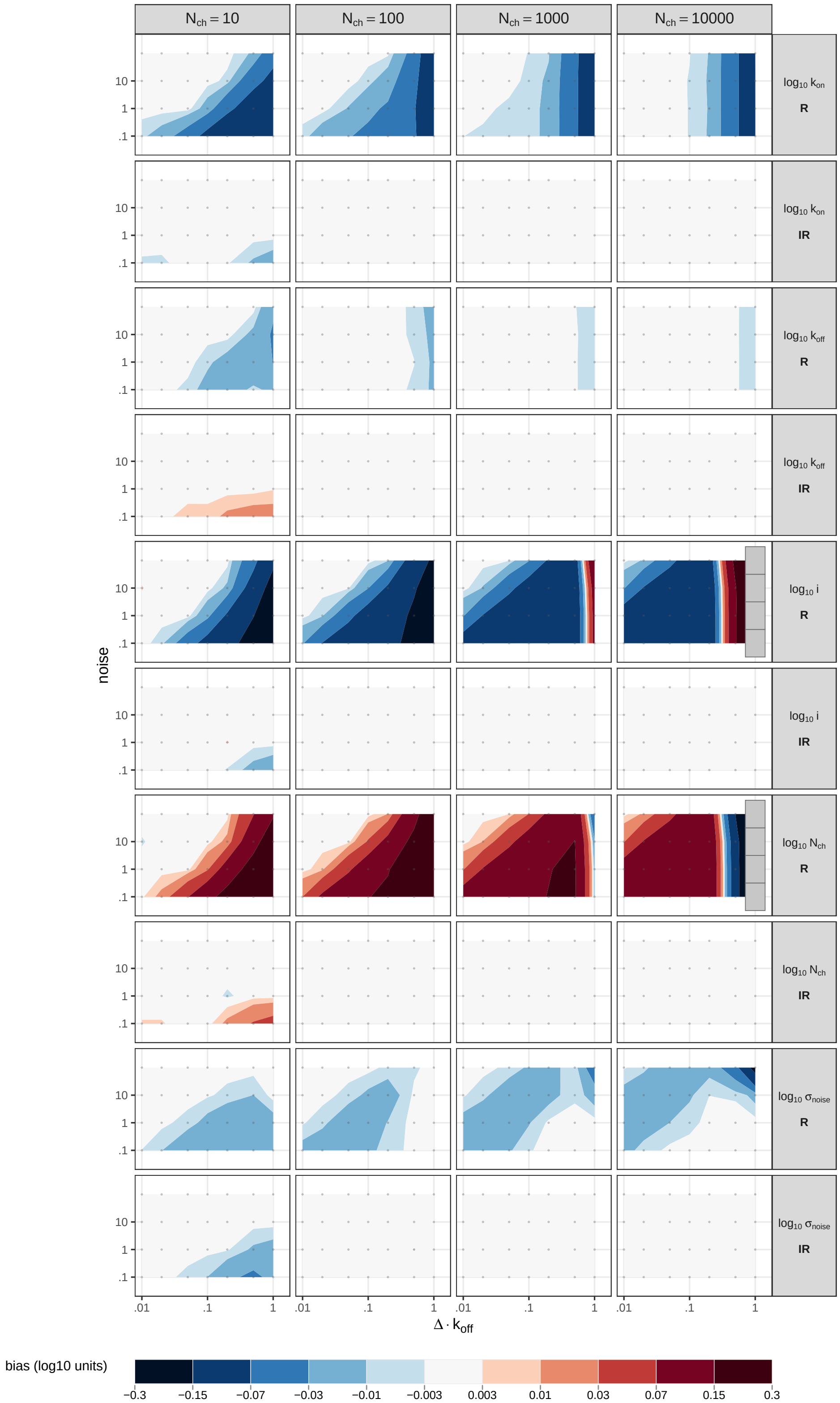


Figure 4—figure supplement 1. The distortion-induced bias for all five parameters, the recursive instantaneous filter against the boundary-conditioned one: the departures of the first are spread across parameters, those of the second are confined.

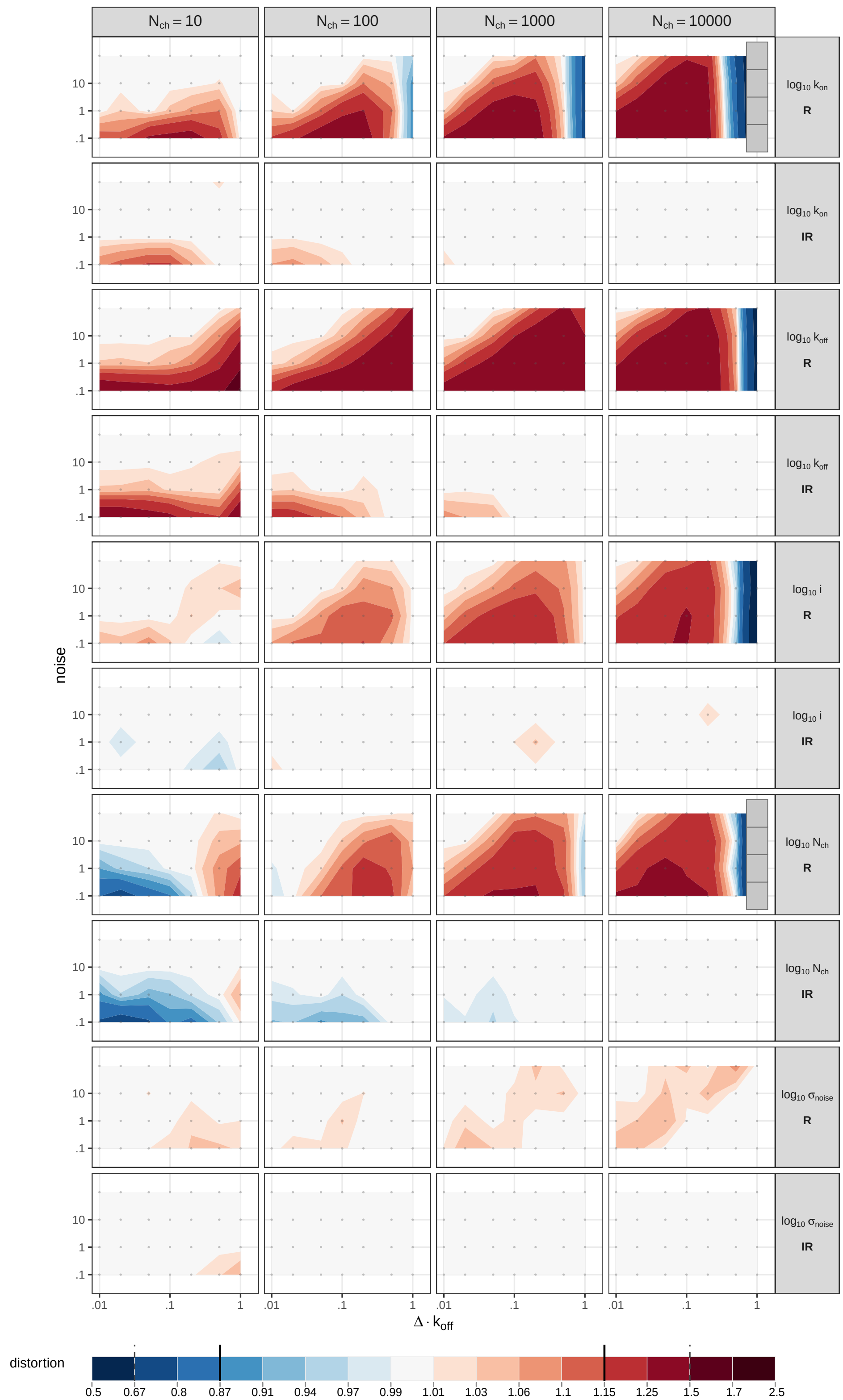


Figure 4—figure supplement 2. The information distortion for all five parameters, the same pair, showing the same contrast in the second moment.

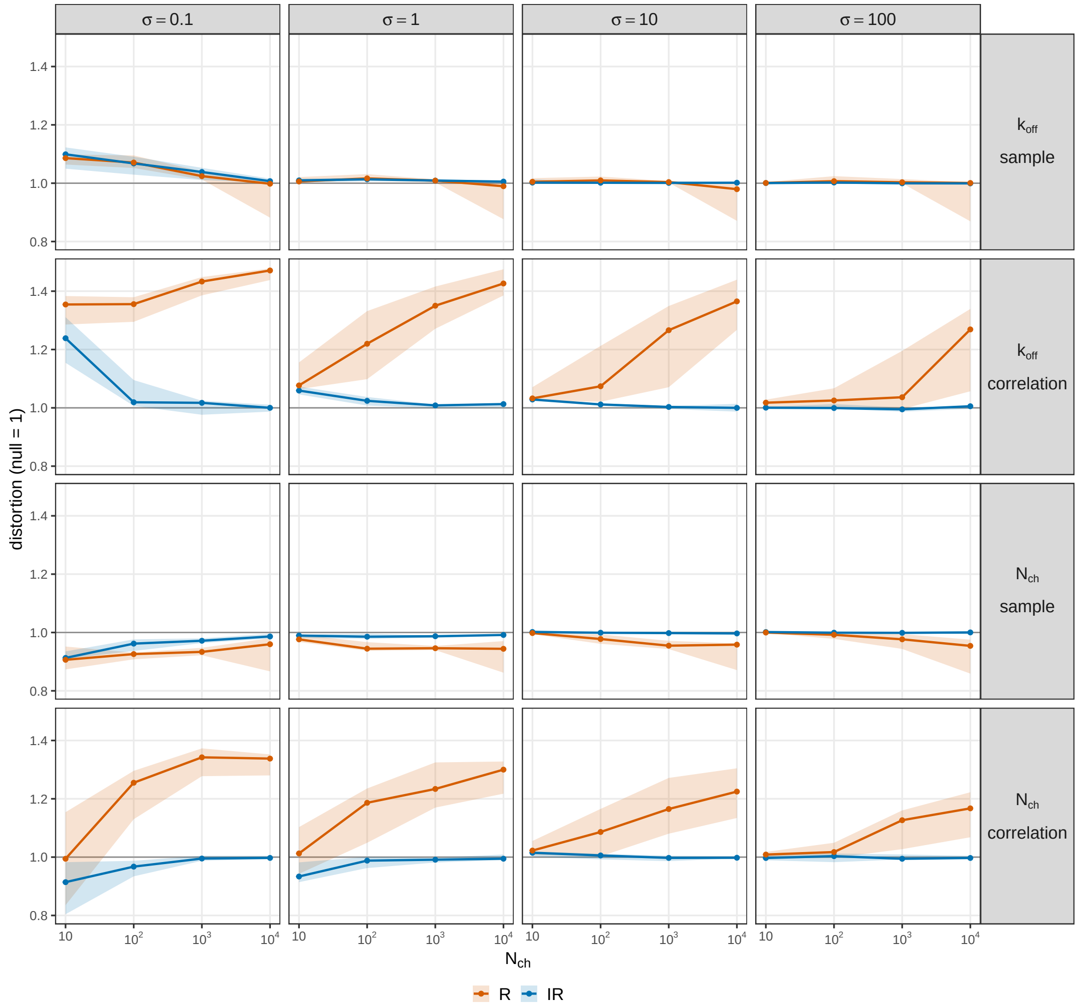


Figure 4—figure supplement 3. The distortion decomposed into a per-sample part and a correlation part, against channel count and resolved by instrumental noise. The per-sample parts of the two recursive members coincide and go to one as channels are added, while the correlation parts separate.

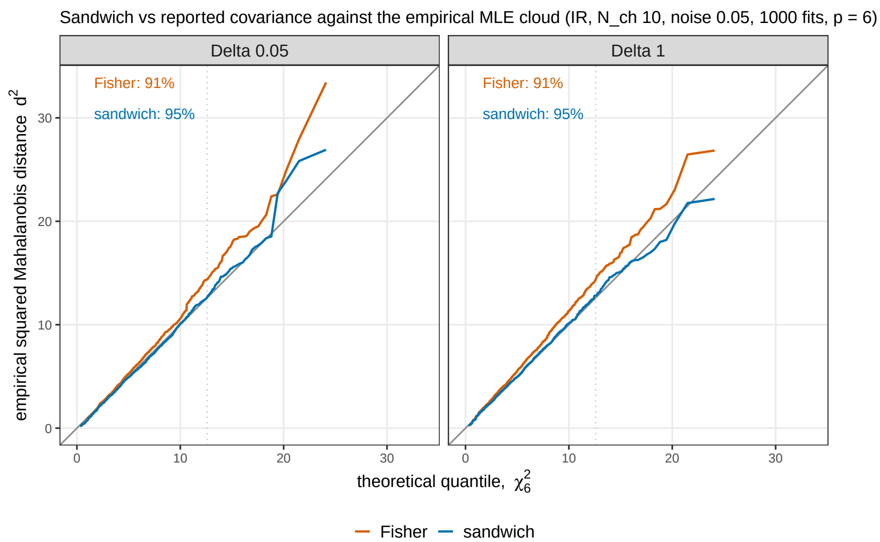
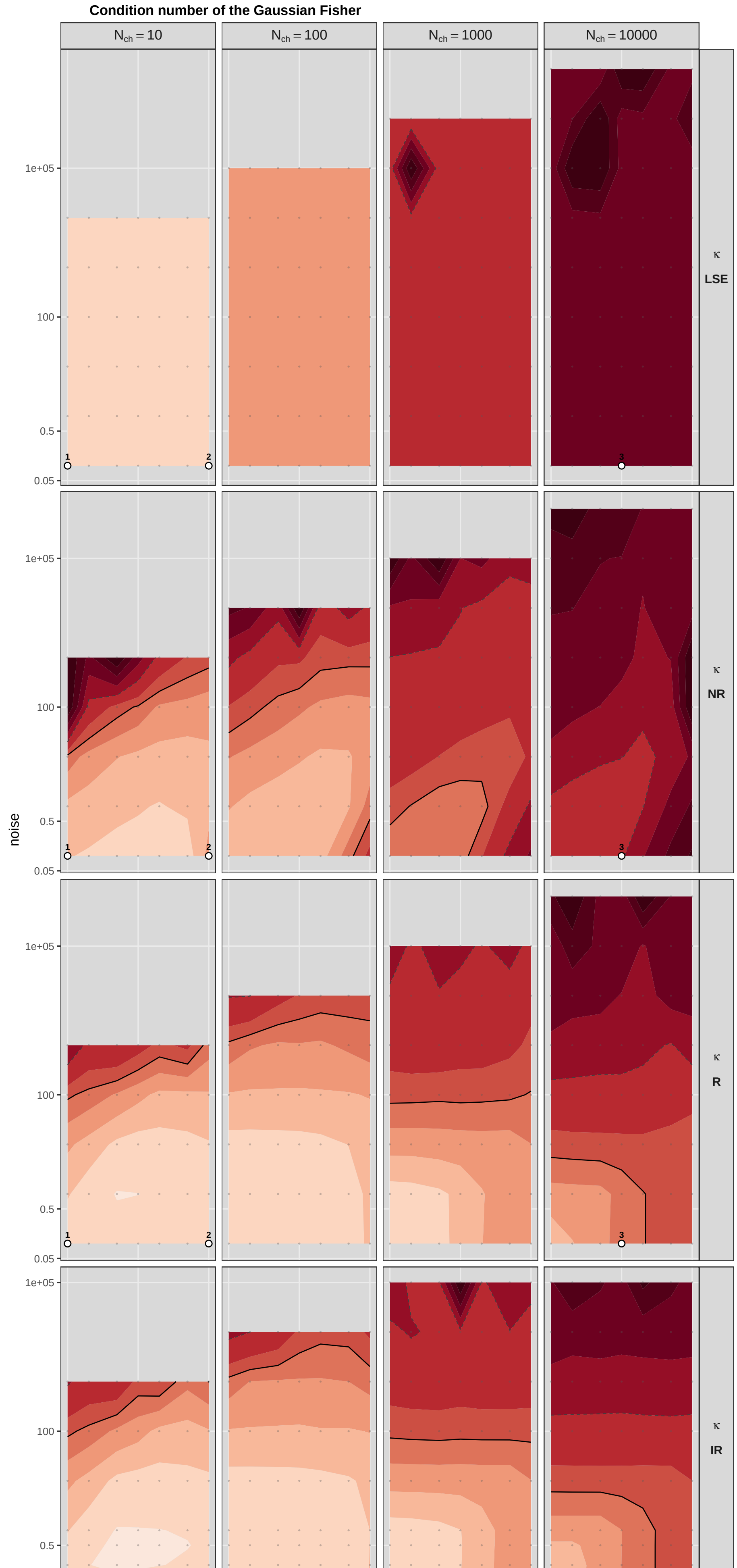
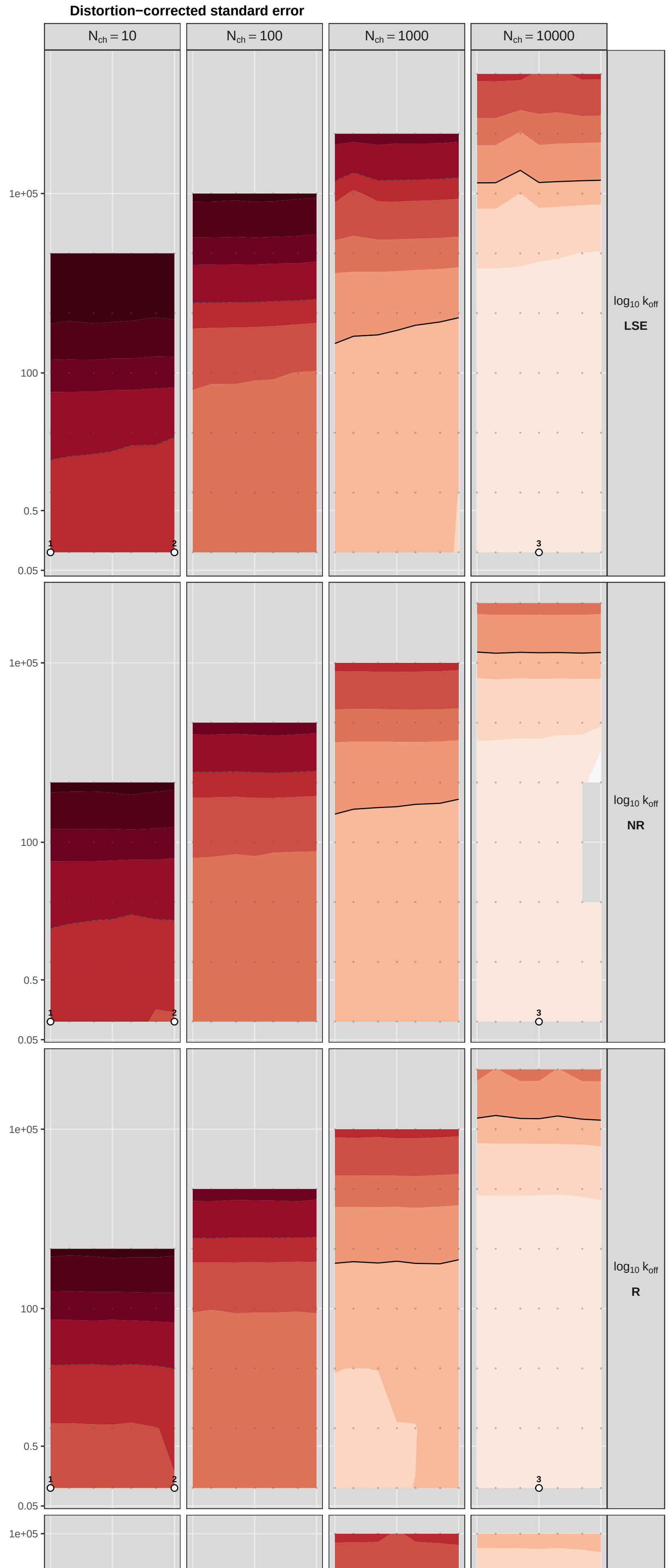


Figure 4—figure supplement 4. The sandwich prediction against the empirical multivariate distribution of the estimates, as a Mahalanobis quantile-quantile plot with a χ^2 reference. The reported Fisher covariance under-covers the nominal 95% region and the sandwich covers it.



Condition number of the Gaussian Fisher
$N_{ch} = 10$
$N_{ch} = 100$
$N_{ch} = 1000$
$N_{ch} = 10000$
1e+05
100
0.5
0.05
noise
κ
LSE
κ
NR
κ
R
κ
IR
1
2
3



Distortion−corrected standard error
$N_{ch} = 10$
$N_{ch} = 100$
$N_{ch} = 1000$
$N_{ch} = 10000$
1e+05
100
0.5
0.05
$\log_{10} k_{off}$
LSE
NR
R
1
2
3

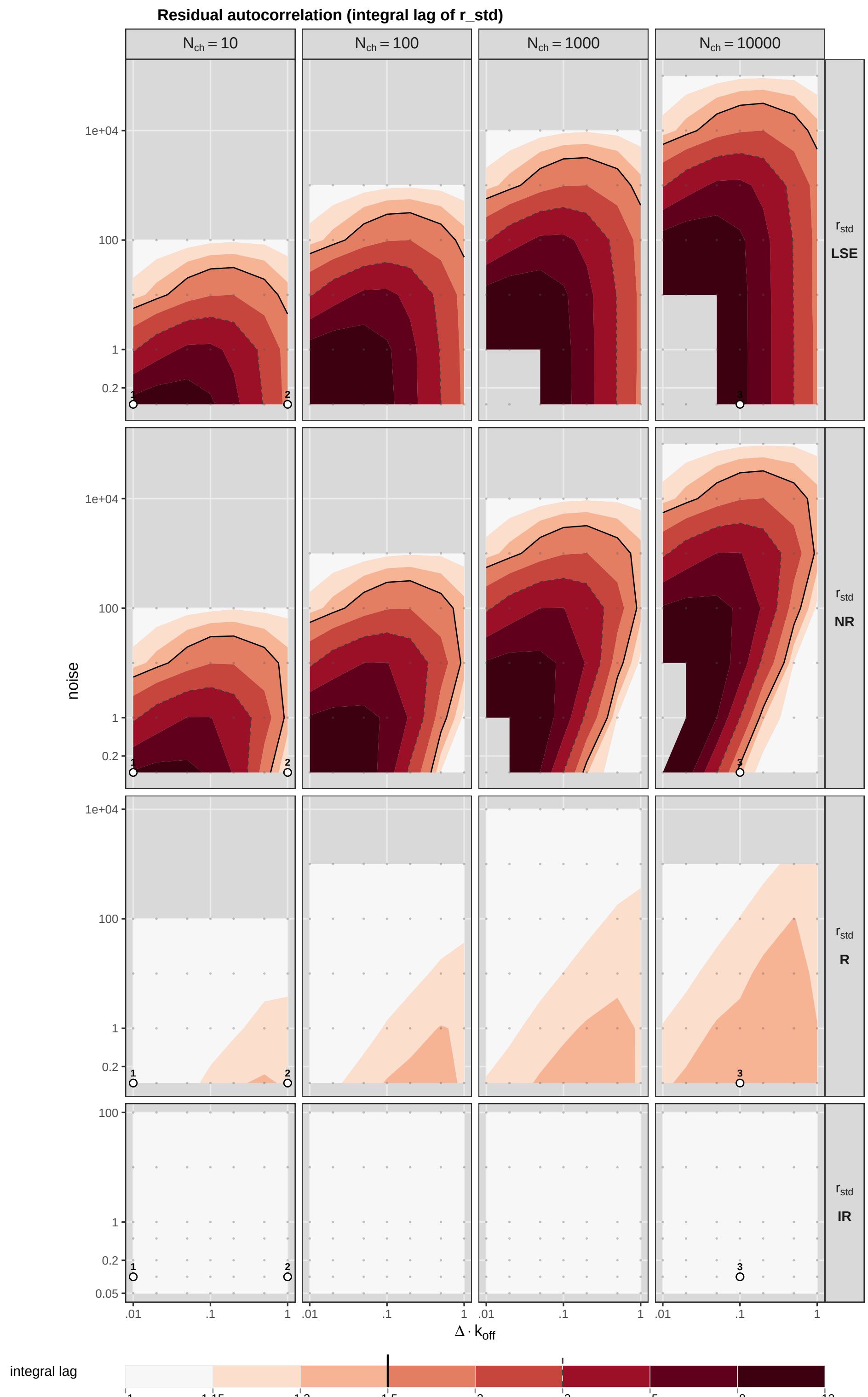
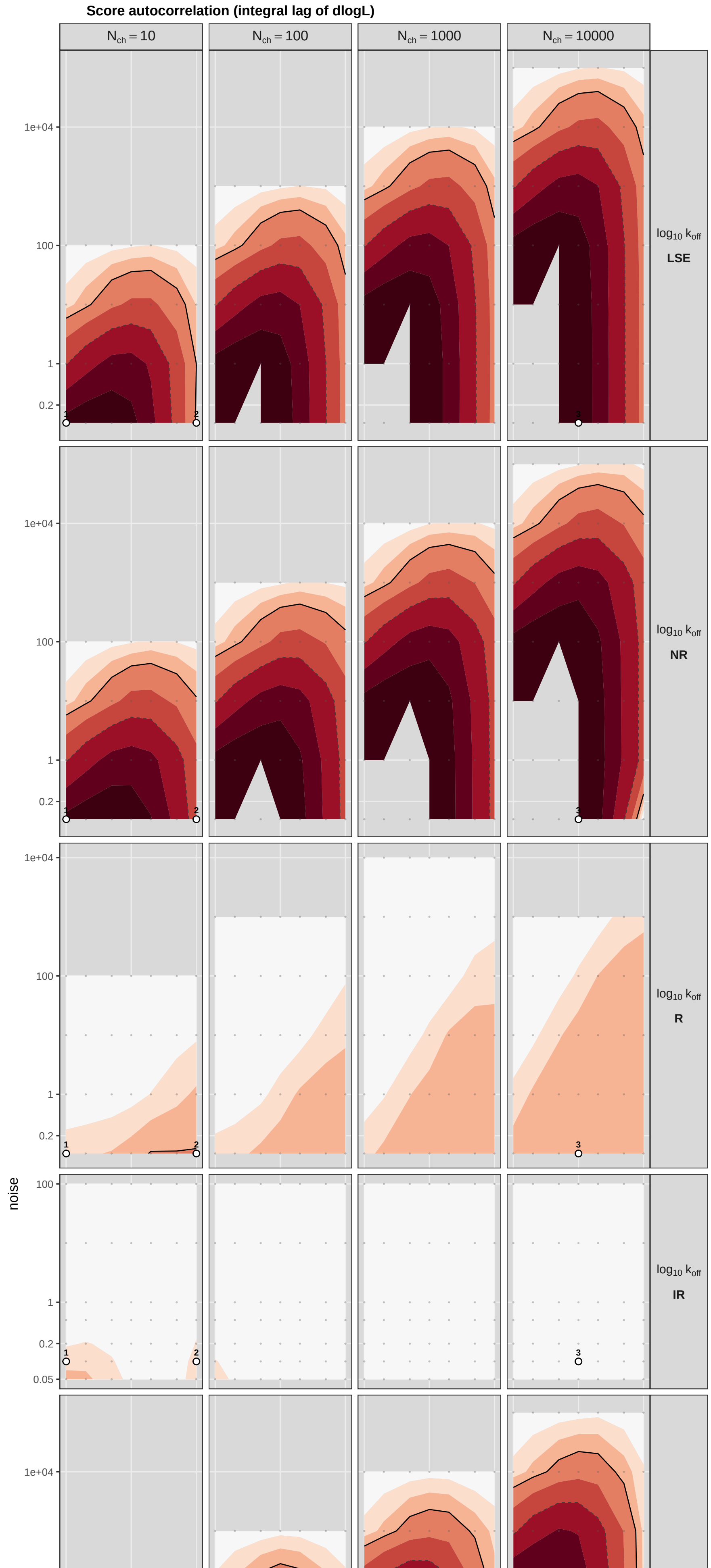
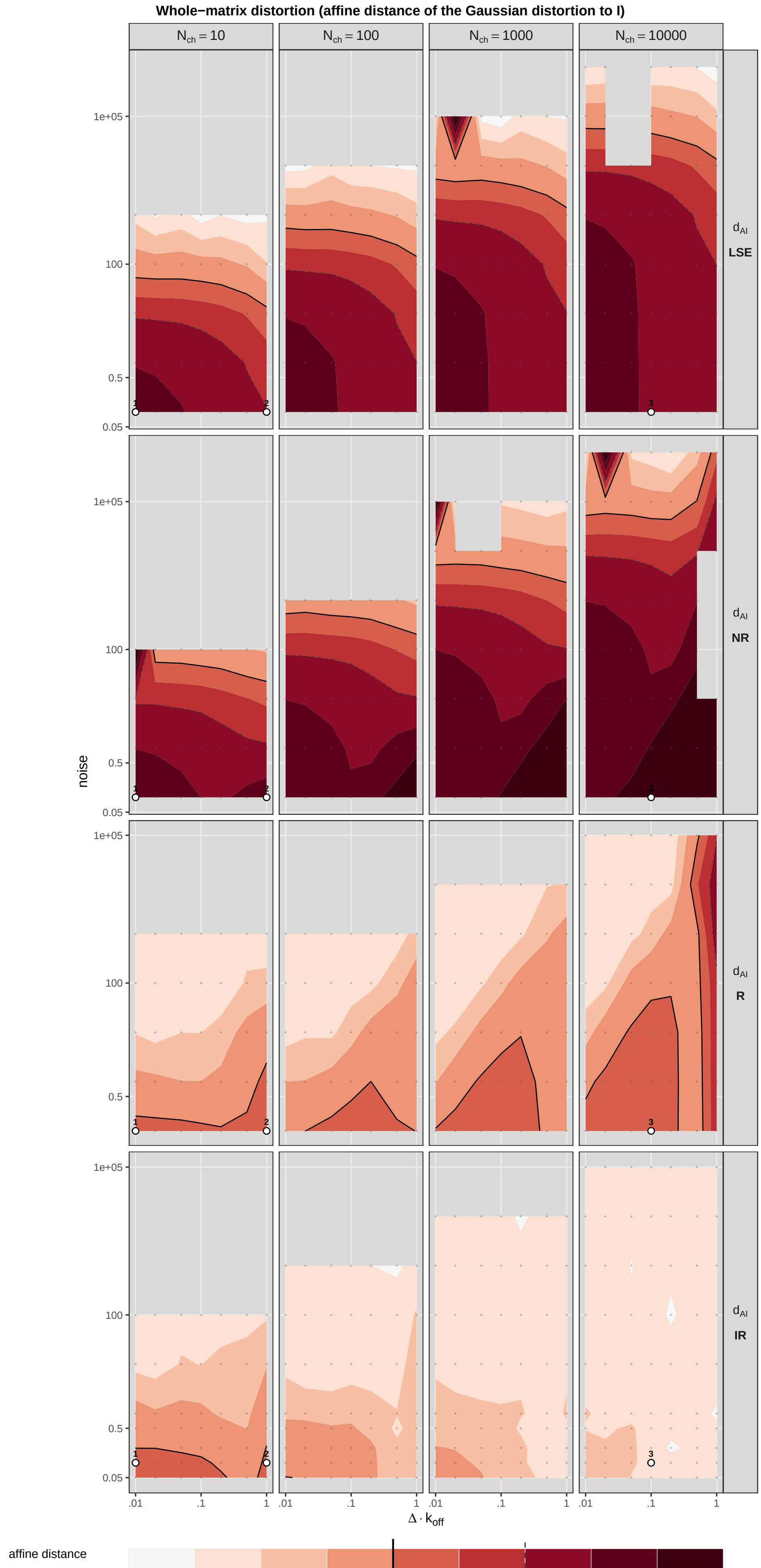
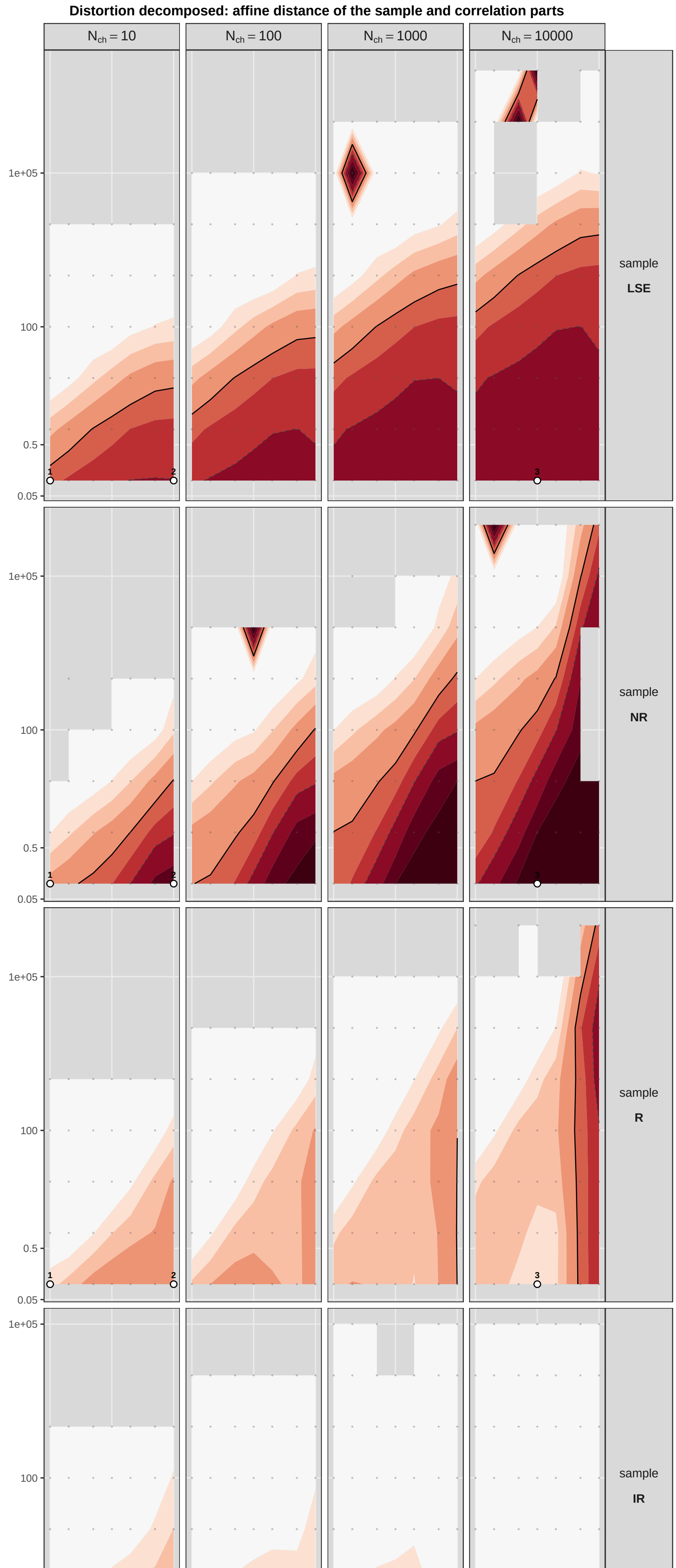


Figure 4—figure supplement 7. The integrated autocorrelation of the standardized residual over the plane: the gating memory that least squares and the open-loop member leave in the data.


Score autocorrelation (integral lag of dlogL)
N_ch = 10
N_ch = 100
N_ch = 1000
N_ch = 10000
log10 k_off
LSE
log10 k_off
NR
log10 k_off
R
log10 k_off
IR
noise
1e+04
100
1
0.2
0.05



Whole−matrix distortion (affine distance of the Gaussian distortion to I)
$N_{ch} = 10$
$N_{ch} = 100$
$N_{ch} = 1000$
$N_{ch} = 10000$
d_{AI} LSE
d_{AI} NR
d_{AI} R
d_{AI} IR
noise
1e+05
100
0.5
0.05
.01
.1
1
$\Delta \cdot k_{off}$
affine distance


Distortion decomposed: affine distance of the sample and correlation parts
$N_{ch} = 10$
$N_{ch} = 100$
$N_{ch} = 1000$
$N_{ch} = 10000$
sample
LSE
sample
NR
sample
R
sample
IR
1e+05
100
0.5
0.05
1
2
3

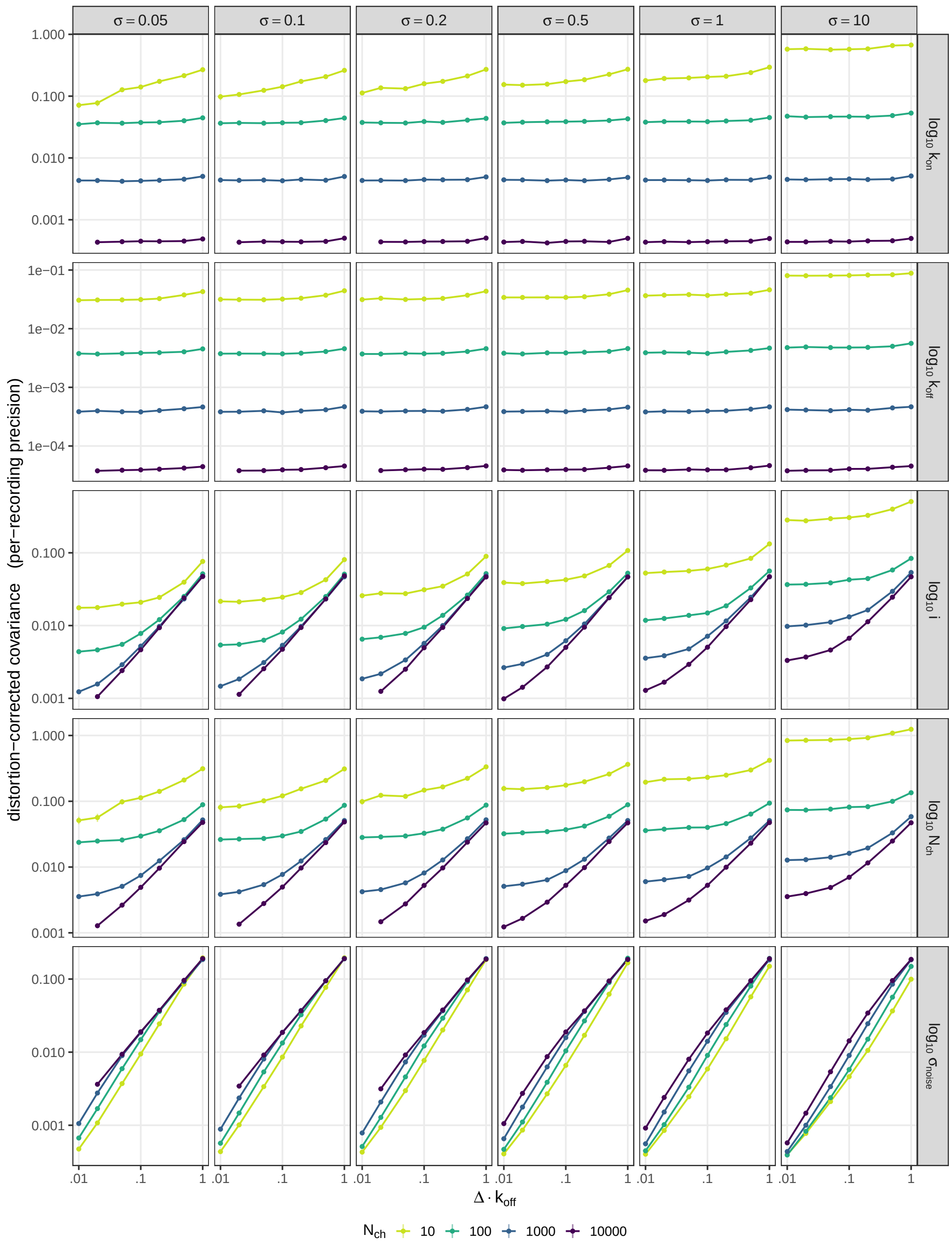


Figure 5—figure supplement 1. The per-recording precision behind the same trade-off, the corrected variance of each parameter in a single recording with the channel-count scaling removed. A larger patch always measures better before pooling, so this view orders the opposite way.

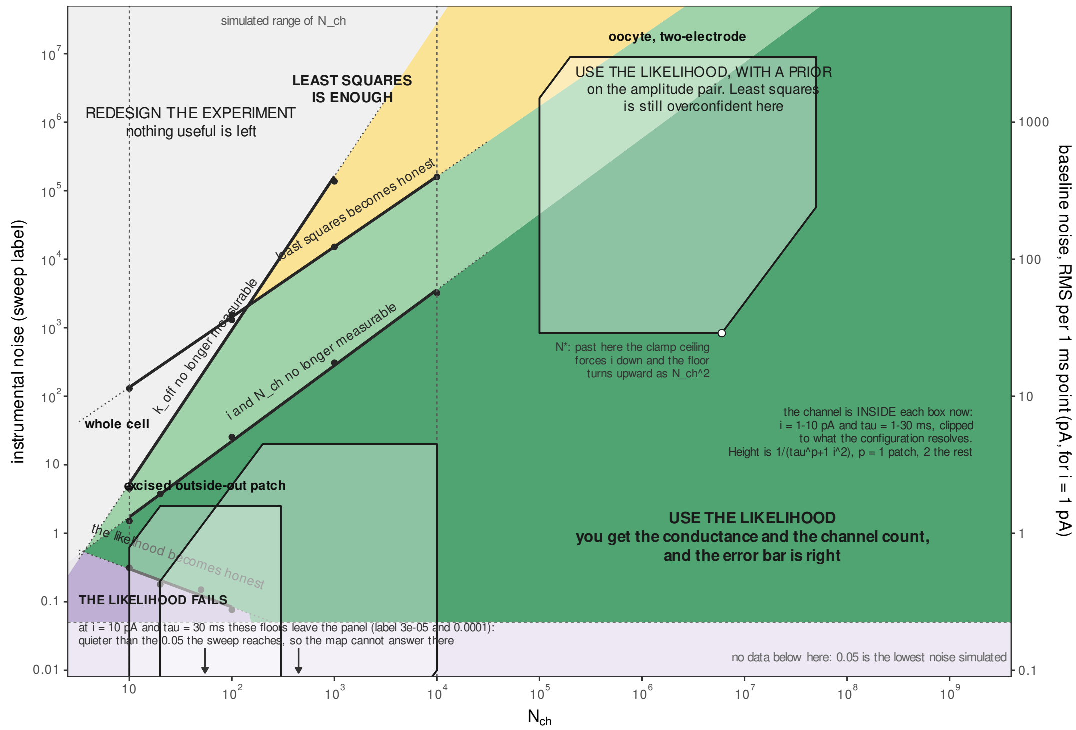


Figure 6—figure supplement 1. The same plane with the five regions shaded and named, and the three recording configurations drawn as footprints, which are provisional orientation rather than measurement, since the noise ranges are read off published traces and the minimum resolvable current for two of the three is an unsourced working value: the closure floor, the two regions in which the likelihood buys something least squares does not, the region in which least squares is at par, and the region in which nothing useful is left. The map is a concept map, not a phase diagram. Its boundaries are level sets of continuous diagnostics, so a looser criterion moves each of them by up to a decade in noise without changing the layout.

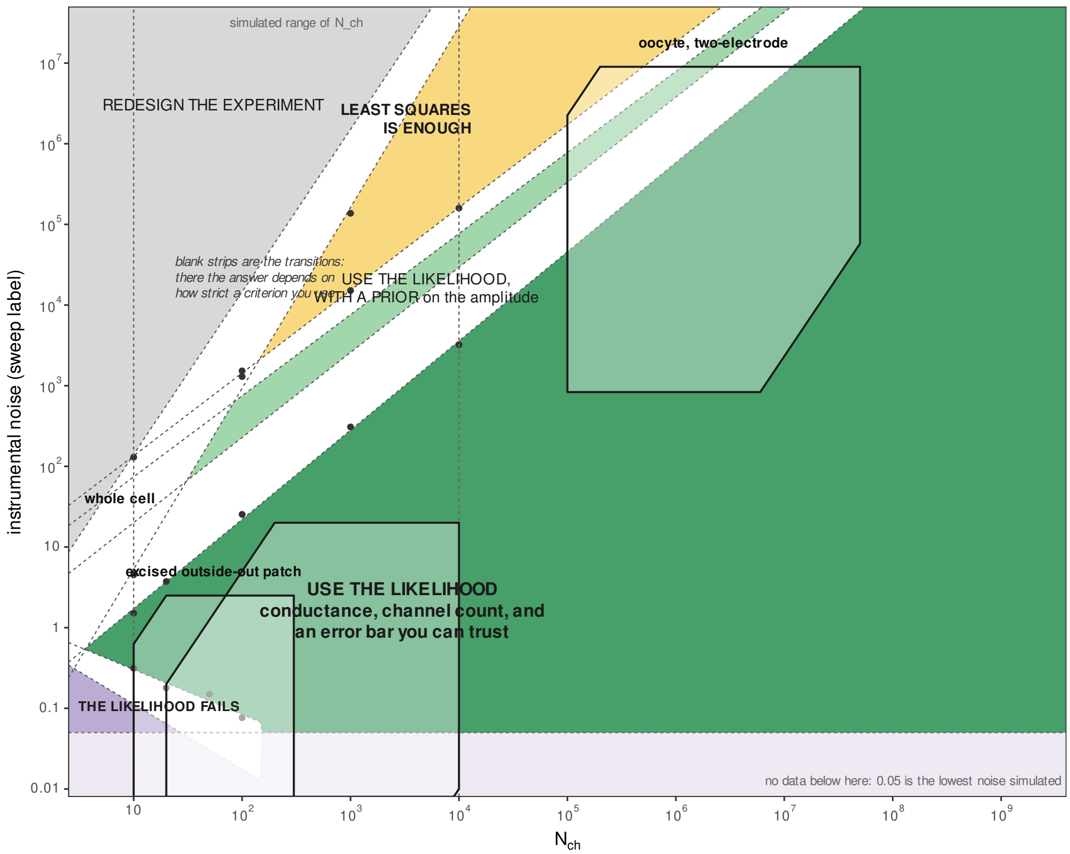


Figure 6—figure supplement 2. The same boundaries drawn as frontier zones rather than lines, which shows directly how much of the plane the choice of criterion moves. The map is a concept map, not a phase diagram. Its boundaries are level sets of continuous diagnostics, so a looser criterion moves each of them by up to a decade in noise without changing the layout.